

Horizon Instability as an Admissible Reconstruction Obstruction

No Uncertified Permanent Horizons under Standing-Conserving Exterior Reconstruction

AASC Black-Hole Endpoint Arc, Paper I: Exterior No-Totalization

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Abstract

Black-hole horizons are ordinarily discussed through classical causal structure, semiclassical evaporation, quantum information, holography, complementarity, and remnant proposals. This paper does not modify general relativity, compute Hawking radiation, derive a Page curve, choose a quantum-gravity model, or assert that classical event horizons cannot be useful idealizations. It proves a narrower fixed-domain closure result. Once an exterior black-hole domain is treated as a coherent construction of persistent physical reference, the admissibility kernel is instantiated automatically: admissibility, standing, reference, and irreversibility become non-optional constraints on same-scope exterior reconstruction.

The standing-sink insight is formulated here as a universal obstruction theorem. The argument proceeds through: kernel instantiation, boundary-trace fixation, exterior witness completion, partial-domain obstruction, conservative hidden-sector collapse, no admissibility-generating structure, route exhaustion, and quotient uniqueness. Because the framework is constraint-theoretic, closure is not an added rhetorical aim; it is the expected result of exhausting all same-scope admissibility-preserving continuations. A permanent horizon is therefore not characterized merely as a region from which signals fail to escape. It is characterized, relative to a declared exterior reconstruction scope, as an attempted continuation of standing-bearing exterior reference after the admissible standing-transport operator has become partial. If no standing-transport certificate exists, totalizing that operator across the old uncertified horizon carrier is inadmissible. Equivalently, the internal horizon totalization obstruction is nonzero.

The finite horizon reconstruction category $\mathcal{C}_{\text{HOR}}$ and its partial-domain refinement $\mathcal{C}_{\text{HOR}}^\partial$ make the comparison class explicit. Objects contain a fixed exterior scope, horizon boundary traces, standing-bearing exterior reference classes, exterior witness algebras, admissible continuation, standing-transport status, certificate data, leakage status, and an envelope ledger for physical inputs not derived by the theorem. Morphisms preserve anchors, exterior witnesses, boundary typing, continuation, transport/certificate status, skin covariance, no-selector discipline, and no-repair history. The quotient $\mathcal{Q}_{\text{HOR}}^\partial$ is universal for admissible same-scope horizon reconstruction: every witness-preserving reconstruction factors through it up to bookkeeping-equivalent redescription.

Three recurring objection classes receive separate treatment. First, global retention is not exterior reconstruction: a globally existing state, correlation, or code preserves exterior standing only when it induces a same-scope certificate on the declared exterior witness algebra. Second, stable remnants are not automatic rescues: a remnant preserving standing must expose an admissible reconstruction map to exterior witnesses, otherwise it is a compression sink, hidden index, or scope transfer. Third, authority cannot generate reconstruction: a duality claim, expert reconstruction procedure, Page-curve reconstruction claim, remnant index, or institutional declaration changes no admissibility status unless it factors through the old exterior witnesses or supplies a certificate; if it does change admissibility without that basis, it is repair, selector

import, carrier laundering, or scope change. A finite toy collapse witness makes the obstruction explicit by displaying leakage, certification, and failed totalization cases on a small reference set.

The analysis also uses the admissible-interior closure discipline: admissibility constraints are necessary conditions for lawful reconstruction, not sufficiency guarantees for persistence or realizability. The paper therefore does not assume that exterior reconstruction must continue indefinitely. It distinguishes three cases: certified continuation, larger-scope continuation, and exhaustion of admissible exterior successors under fixed constraints. Exhaustion is not a process, entropy drain, dynamical decay, or failure of the admissibility rules; it is the absence of any standing-preserving successor object in the declared scope. A model may acknowledge exhaustion, but it may not then present an uncertified permanent horizon as a continuing same-scope standing-preserving reconstruction object.

The main theorem is conditional and scope-locked. In a fixed witness-complete exterior scope requiring standing-bearing reference preservation, an uncertified permanent horizon is inadmissible as a continuing same-scope reconstruction object. Any admissible horizon model must instantiate leakage, evaporation, certified exterior reconstruction, holographic anchor transport, complementarity with redescription-stable correspondence, a non-sink quotient structure, or explicitly classify the case as exhaustion of exterior reconstruction rather than as standing-preserving permanence. Hawking radiation and related mechanisms are possible dynamical realizations of the structural requirement, not assumptions of the proof.

AASC Black-Hole Endpoint Arc Position

Shared arc spine: kernel/AMetric invariant $\rightarrow$ licensed AASC-GR black-hole regime $\rightarrow$ exterior no-totalization $\rightarrow$ standingless residual interior $\rightarrow$ governance-depth exhaustion $\rightarrow B_A^ \rightarrow$ AMetric homology $\rightarrow$ fresh internal governed domain.*

Shared series abstract. The AASC Black-Hole Endpoint Arc is a five-manuscript sequence that treats black holes as a fixed-domain stress test for AASC. It first role-types the licensed black-hole regime, then proves the exterior no-totalization obstruction, then classifies the uncertified horizon-isolated carrier as residual coherence rather than ordinary standing-bearing interiority, then exposes the endpoint AMetric normal form and no-crossing homology, and finally classifies the counter-boundary survivor as fresh internal governed-domain status. The sequence is conditional and one-way: later endpoint or capstone conclusions never authorize earlier target formation.

Paper label. Paper I / Exterior no-totalization. This manuscript proves the fixed-exterior-scope no-totalization result for uncertified permanent horizons.

This manuscript is Paper I of the proof sequence after the black-hole role-classifier. Its task is exterior and fixed-domain: it proves that an uncertified permanent horizon cannot be retained as a continuing same-scope standing-preserving exterior reconstruction object.

Imported into this paper. The paper imports the licensed black-hole target discipline: fixed exterior scope, standing-bearing exterior reference classes, exterior witness algebra, continuation relation, boundary-trace typing, certificate status, and no-selector/no-repair discipline.

Proved here. The paper proves the exterior no-totalization result. In the uncertified permanent-horizon case the standing-transport operator is partial, the threatened class lacks an admissible future image in the old exterior witness algebra, and any same-scope totalization requires certificate, leakage, non-sink quotient, repair, selector import, carrier laundering, bookkeeping, or scope change.

Exported to the next paper. The export is the standing break for crossed content, the horizon-isolated carrier E_{int} as an old-scope domain where standing transport is undefined, and the classification of uncertified permanence as exterior-successor exhaustion rather than successful standing-preserving continuation.

Not claimed here. This paper does not yet classify the interior carrier as residual coherence, does not expose a New AMetric Boundary, does not prove AMetric homology, and does not assert any fresh internal governed domain. It only closes the exterior reconstruction route.

Series dependency lock. The only arc input used here is the licensed target package exported by the role-classifier. The theorem does not import residual coherence, governance-depth, AMetric homology, or counter-boundary universe-status. Its export is the old-scope standing break and operator-domain fact

$$E_{\text{int}} \cap \text{dom}(T_H) = \emptyset$$

for every threatened class left uncertified by leakage, certificate, or non-sink quotient.

Forward handoff. The next paper may ask what kind of interior status remains for crossed content after this standing break. That question is deliberately not answered here. This paper supplies only exterior no-totalization and exterior-successor exhaustion; it does not classify the hidden carrier as ordinary interior, residual coherence, nullity, or New AMetric Boundary.

Certificate terminology lock. The word *certificate* has four arc-level uses, and the uses are not interchangeable. A *standing-transport certificate* is a same-scope map or witness assignment that preserves a threatened reference class in the exterior reconstruction problem. An *exterior witness certificate* is a concrete exterior, boundary, radiation, remnant, holographic, island, or quotient map that removes a burden from the unresolved-residue ledger. An *endpoint-certified target* is not a rescue certificate; it is the antecedent package in which a nonempty unresolved residue, old-scope standing loss, residual metric-support exhaustion, closed false endpoints, null-route closure, and no prior re-governance make endpoint AMetric exposure the target. A *capstone-certified endpoint* is the Paper IV input: the homology handoff together with endpoint certification and no-crossing. A horizon can be transport-certified and thereby exit the exterior obstruction; an unresolved residue can be endpoint-certified precisely because no such transport certificate has resolved it.

Exterior-quotient and metric-regime lock. The proof-level object of the arc is the declared exterior standing quotient of the black-hole model, not a particular metric-regime representative. In the familiar settled Kerr–Newman example the quotient is displayed by mass, electric charge, and angular momentum, but that example does not restrict the theorem to stationary representatives and does not make absence of hair a separate proof premise. Hair-like, dynamical, evaporating, ringdown, perturbative, semiclassical, quasi-local, high-curvature, or quantum-gravity structure has same-scope force only when it is declared as exterior quotient content, an exterior witness, a leakage channel, a certificate, a non-sink quotient, or richer governance. Such structure subtracts from the unresolved residue exactly to the extent that it re-fixes standing-bearing content. If it does not re-fix standing, it is skin, bookkeeping, hidden load, repair, selector import, carrier laundering, or scope change; it does not provide an additional endpoint route.

Exterior-quotient subtraction lock. The exterior obstruction is quotient-based rather than based on absence of hair or stationarity. Let the declared exterior quotient and witness package include every exterior parameter ledger, hair-like witness, leakage record, radiation correlation,

perturbative record, boundary map, holographic or island reconstruction map, remnant map, non-sink quotient, and richer-scope certificate admitted by the model. Those structures reduce the threatened burden only to the extent that they re-fix standing-bearing content in the old exterior reconstruction scope. The residue left after that subtraction is the object tested by the partial-domain theorem. Hair-like or nonstationary structure that does not re-fix standing does not alter the partiality of the old standing-transport operator.

Referee-facing arc firewall. The exterior theorem is deliberately insensitive to the preferred interior story. Classical event-horizon exactness, local GR regularity, global unitarity, Page-curve behavior, remnant capacity, holographic encoding, island reconstruction, complementarity, final-state projection, or a quantum-gravity completion affects this paper only when it induces one of the admitted exterior routes: leakage, evaporation, a standing-transport certificate, a non-sink quotient, explicit successor exhaustion, or a declared scope change. Without that induced route, the appeal is a global-retention or authority claim rather than a same-scope exterior reconstruction witness.

This firewall prevents two common misreadings. First, the theorem does not assume that the interior is empty, standingless, AMetric, or universe-generating; those are not exterior conclusions. Second, the theorem does not infer the capstone result from no-totalization. It exports only the standing break and partial-domain obstruction. A same-scope objection must therefore either supply the missing exterior certificate or show that the partial-domain classification of T_H is wrong under the declared witness algebra.

Pass 6 notation lock. The local scope symbol $\mathcal{S}_{\text{hor}}$ is this paper’s exterior-reconstruction specialization of the Paper 0 target package $\mathcal{P}_{\text{BH}}^0$. The partial operator $T_H : X_H \multimap R_{\text{ext}}$ is an old-scope standing-transport assignment, not an interior-governance operator. The carrier E_{int} means only the horizon-isolated old-scope carrier on which T_H is undefined for the threatened class; it is not yet a residual-coherence regime and not the AMetric endpoint B_A^* . The symbols $\text{Succ}_{\mathcal{S}_{\text{hor}}}$ and $\text{Exh}_{\mathcal{S}_{\text{hor}}}$ denote exterior successor availability and exterior reconstruction exhaustion only. This paper does not use R_H , B_A^* , or D_1 as theorem conclusions.

Pass 9 local referee-response lock. For local review, this paper has one theorem target: exterior same-scope no-totalization. It does not prove that the interior is empty, residual, AMetric, or universe-generating. The relevant hostile alternative is therefore not “information exists somewhere globally” but “the declared exterior witness algebra or certificate package already re-fixes the threatened class.” A global state, unitary evolution, dual description, Page-curve behavior, remnant label, or reconstruction algorithm counts against the theorem only when it induces a standing-transport certificate on the old exterior scope. Otherwise it is larger-scope retention, bookkeeping, hidden selector import, or an unlicensed repair of the partial operator. The only export to Paper II is the old-scope standing break and undefined transport on the horizon-isolated carrier; no interior-status conclusion is exported here.

Pass 10 hostile-referee closure lock. The strongest hostile counterexample to this paper is not the existence of global information, a unitary completion, a Page curve, a holographic dual, an island prescription, a remnant Hilbert space, or an expert reconstruction algorithm. The counterexample must be a same-scope exterior standing-transport certificate for the threatened reference class, or else a declared change of target. If a standard program supplies that certificate, the theorem’s

uncertified antecedent is false and the program is an allowed resolution. If it does not, the program is larger-scope retention, bookkeeping, selector import, repair, or carrier laundering relative to the old exterior problem. This paper therefore cannot be refuted by storage or calculability alone; it is refuted only by old-scope re-fixability.

Main sequent. On the declared horizon domain,

$$\begin{aligned} \text{Adm}(\mathcal{S}_{\text{hor}}), \quad \text{Stand}(\mathcal{R}_{\text{ext}}), \quad \text{Permanent}(B_H), \quad \text{Cert}_H = \emptyset \\ \implies \neg \text{Adm}_{\mathcal{S}_{\text{hor}}}(\text{PermHorizon}) \wedge O_H(H^\partial) \neq 0. \end{aligned} \tag{0.1}$$

The reconstructive non-guarantee principle is

$$\text{Succ}_{\mathcal{S}_{\text{hor}}}(H^\partial) = \emptyset \implies \text{Exh}_{\mathcal{S}_{\text{hor}}}(H^\partial) \text{ not constraint failure or admissible permanence.} \tag{0.2}$$

The constructive alternative is

$$\begin{aligned} \text{Adm}(\text{continuing horizon model}) \implies & \text{leakage} \vee \text{evaporation} \\ & \vee \text{certified exterior reconstruction} \vee \text{non-sink quotient} \\ & \vee \text{explicit exterior-successor exhaustion.} \end{aligned} \tag{0.3}$$

Dependency note. The present formulation preserves the original target load: standing conservation excludes permanent standing sinks. The argument does not introduce a new conservation principle, and it does not assume that admissible reconstruction must be realizable indefinitely. It uses the AASC obstruction architecture developed in the baryogenesis and neutrino reconstruction papers: fixed-domain scope lock, boundary-trace fixation, partial-domain closure, conservative reconstruction collapse, anchor/tensor/skin role typing, admissibility-generating-structure impossibility closure, and finite route exhaustion.

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1 Introduction: the theorem target

Classical general relativity permits spacetimes containing event horizons. Semiclassical field theory on curved spacetime gives Hawking radiation and black-hole thermodynamics. Quantum-information approaches ask how exterior observers recover information, whether horizon evolution is unitary, whether holography encodes the interior, and whether complementarity avoids inconsistent duplication. None of those programs is replaced here.

The target question is prior and narrower:

Can a permanent horizon be an admissible same-scope terminal boundary for standing-bearing exterior reconstruction?

The answer proved here is no, unless the model supplies a standing-transport certificate. A permanent horizon without such a certificate acts as a standing sink: standing-bearing reference crosses or becomes hidden behind a boundary, while no future admissible exterior description contains a redescription-stable image of the reference. That is not merely informational inconvenience. It is an inadmissible loss of standing on the declared exterior scope.

The original note expressed this in three steps:

- (a) standing is conserved under admissible conditioned evolution;
- (b) permanent horizons are standing sinks relative to exterior scope unless compensating mechanisms exist;
- (c) no permanent standing sink can arise in admissible evolution.

That argument is correct as a consequence note, but too compressed for a mathematical-physics presentation. Natural questions include: What is the comparison class? What counts as an admissible exterior image? Is holography a counterexample? Does global unitarity suffice? What about remnants, infinite-time recovery, observer-relative horizons, complementarity, final-state repair, or baby universes? The purpose of this paper is to classify those routes at the formal level.

The strengthened claim is not that every formal spacetime containing a classical event horizon is inconsistent. The claim is that an *uncertified permanent horizon* is not an admissible same-scope reconstruction object when the exterior domain requires standing-bearing reference preservation. Classical event horizons remain classical idealizations; the theorem concerns whether such horizons can serve as terminal standing-preserving boundaries in a fixed exterior reconstruction problem.

Reconstruction-only scope restriction. The theorem is reconstructive rather than ontological. It classifies admissible same-scope exterior reconstruction structures under fixed standing-preserving continuation burdens. It does not assert that inaccessible interiors, global states, or spacetime regions are ontologically nonexistent, nor does it require global nonexistence of inaccessible states. A failure of exterior admissible reconstruction is a failure of lawful reconstruction availability on the declared scope, not a claim that the larger physical or mathematical object has ceased to exist. The conclusion is not that horizons are dynamically forbidden; it is that uncertified permanent horizon reconstructions are not admissible same-scope exterior reconstructions.

Claim. In a fixed exterior horizon reconstruction domain, standing-bearing reference may be lost behind a candidate horizon only if some admissible mechanism transports, leaks, reconstructs, or

quotients that standing without creating a standing sink. A horizon that permanently severs such transport without certificate is excluded by standing conservation and partial-domain closure. The admissible alternatives are leakage, evaporation, certified holographic/boundary reconstruction, complementarity with explicit anchor correspondence, or a non-sink redescription.

Why the obstruction form matters. Readers interested only in the main obstruction may track the theorem chain: kernel instantiation, partial-domain obstruction, acquisition exhaustion, successor exhaustion, and no uncertified permanent horizon.

The bare standing-sink theorem proves that permanent standing loss is inadmissible. The universal-obstruction formulation proves more: every same-scope route by which an uncertified permanent horizon might be retained collapses into one of four types: bookkeeping, hidden selector import, post-hoc repair, or scope change. If a route supplies a genuine standing-transport certificate, it is not a counterexample; it is one of the admissible non-sink realizations demanded by the theorem.

2 Scope lock and non-claims

The theorem is conditional and fixed-domain. A comparison is same-scope only when it preserves the exterior reconstruction target. Changing the target may be useful physics, but it is not a same-scope rescue of an uncertified permanent horizon.

Definition 2.1 (Exterior horizon reconstruction scope). A fixed exterior horizon reconstruction scope is a tuple

$$\mathcal{S}_{\text{hor}} = (\mathcal{E}_{\text{ext}}, B_H, \mathcal{R}_{\text{ext}}, \mathcal{W}_{\text{ext}}, \text{Cont}, \text{Env}), \quad (2.1)$$

where:

- (i) $\mathcal{E}_{\text{ext}}$ is the exterior admissible description domain;
- (ii) B_H is a candidate horizon or horizon-like boundary interface;
- (iii) $\mathcal{R}_{\text{ext}}$ is the family of standing-bearing exterior reference classes required by the scope;
- (iv) $\mathcal{W}_{\text{ext}}$ is the family of exterior witnesses, records, fields, correlations, detector data, boundary functionals, or asymptotic observables admitted by the scope;
- (v) Cont is the admissible continuation relation for standing-bearing exterior descriptions;
- (vi) Env is the envelope ledger: geometric approximation, semiclassical regime, quantum assumptions, boundary conditions, and numerical/dynamical inputs not derived by this theorem.

Definition 2.2 (Same-scope horizon competitor). A proposal P is a same-scope competitor to the present theorem only if it preserves $\mathcal{S}_{\text{hor}}$ up to admissible redescription. In particular, it must preserve the exterior witness family, the standing-bearing reference burden, the boundary typing of B_H , the continuation relation, the transport/certificate status, and the envelope ledger. A proposal that adds a new detector algebra, new asymptotic boundary, hidden branch, baby universe, final-state selector, quantum-gravity sector, or holographic dictionary changes scope unless it induces the same standing-preserving exterior witness object.

Definition 2.3 (Standing-bearing exterior burden). A scope $\mathcal{S}_{\text{hor}}$ carries a standing-bearing exterior burden when at least one reference class $r \in \mathcal{R}_{\text{ext}}$ must remain re-fixable under admissible exterior continuation and redescription. Examples include persistent exterior records, correlations, field

imprints, conserved charges with exterior witnesses, boundary traces, or reconstructible relational data. The burden is absent only if the declared exterior problem no longer requires standing-bearing physical reference.

Definition 2.4 (Formal exterior witness algebra). The declared exterior witness algebra is

$$\mathcal{A}_{\text{wit}} = \text{Alg}(\mathcal{W}_{\text{ext}}, \mathcal{A}_{\partial H}^{\text{adm}}, \text{Cont}, \text{Cert}_H), \quad (2.2)$$

where $\mathcal{A}_{\partial H}^{\text{adm}}$ denotes horizon-boundary traces admitted by the exterior scope. The algebraic notation is schematic: it records closure under the declared exterior reconstruction operations, not a claim that a complete quantum-gravity observable algebra has been constructed.

Definition 2.5 (Horizon carrier typing). A horizon reconstruction carrier is typed into three admissibility roles:

$$X_H = X_{\text{anch}} \sqcup X_{\text{tens}} \sqcup X_{\text{skin}}. \quad (2.3)$$

Here X_{anch} contains exterior and boundary anchors required for reference fixation; X_{tens} contains quotient-level invariant witness content such as correlations, radiation functionals, causal relations, and certificate maps; and X_{skin} contains charts, foliations, labels, coordinate representatives, and verbal redescriptions that alter no standing-bearing witness. A same-scope rival must preserve this role typing up to role isomorphism.

Definition 2.6 (Certificate-complete scope). A horizon scope is certificate-complete when every admitted preservation of threatened standing is represented inside one of the following: $\mathcal{A}_{\text{wit}}$, a leakage channel Leak , a standing-transport certificate Cert_H , or a declared non-sink quotient. The term is comparison-class relative. It does not deny that a larger theory may add new witnesses; it says that such additions are not same-scope unless they factor back through the declared witness package.

Theorem 2.7 (Finite exterior-scope closure theorem). *Within a fixed exterior horizon reconstruction scope, a proposed permanent-horizon reconstruction is a same-scope competitor only if it preserves exterior anchors, standing-bearing reference classes, exterior witnesses, admissible continuation, no-selector discipline, and no-repair discipline. A proposal that adds a new recovery channel or global domain is either bookkeeping-equivalent on the declared exterior witnesses or a larger-domain proposal.*

Proof. The target object is not an arbitrary spacetime description. It is a same-scope exterior reconstruction of standing-bearing reference through a candidate horizon history. If a proposal changes exterior anchors, it changes what is being reconstructed. If it changes witnesses, it changes the evidential domain. If it changes continuation, it changes admissible history. If it introduces a hidden selector or post-hoc repair, it creates standing from outside the declared domain. If it changes none of these, it induces the same reconstruction object and is bookkeeping-equivalent. These are exhaustive by fixed-domain role decomposition: any admissibility-bearing variation must alter anchor content, tensor/witness content, skin, continuation, or scope. No additional same-scope governance locus remains. $\square$

Non-claims

This manuscript does not claim:

- a derivation of Hawking temperature, greybody factors, entropy, or evaporation time;

- a replacement for general relativity, semiclassical gravity, quantum field theory, string theory, loop quantum gravity, or holography;
- a full black-hole information-paradox solution;
- a Page-curve calculation;
- a constructive AQFT model of black-hole evaporation;
- a denial that classical GR admits event-horizon solutions;
- a denial of global unitarity where a model supplies it;
- a proof that holography, complementarity, or remnants are false in larger domains;
- a numerical prediction of radiation channels or microscopic black-hole states.

The claim is narrower and stronger within its scope: an uncertified permanent standing sink is not an admissible exterior reconstruction object. If a physical theory supplies certified exterior standing transport, leakage, or non-sink reconstruction, it satisfies rather than defeats the theorem.

Cosmological and observer-horizon limitation. Cosmological, de Sitter, Rindler, and observer-relative horizons are not classified merely by name in this manuscript. They fall under the theorem only when they are explicitly typed as standing-bearing exterior reconstruction obstructions on a fixed continuation scope. Otherwise they require a separate scope definition, witness algebra, and successor relation.

3 Reconstruction dependence and admissible-successor discipline

The admissible-interior closure papers supply a meta-level guardrail. Admissibility conditions define what would count as a lawful interior or exterior reconstruction; they do not guarantee that a reconstruction continues to be realizable. This distinction prevents the present theorem from becoming an illicit persistence principle.

Definition 3.1 (Exterior admissible successor set). Let H^∂ be a partial horizon reconstruction object in the fixed scope $\mathcal{S}_{\text{hor}}$. The exterior admissible successor set is

$$\text{Succ}_{\mathcal{S}_{\text{hor}}}(H^\partial) = \{K^\partial : H^\partial \rightsquigarrow K^\partial \text{ by a standing-preserving admissible exterior continuation}\}. \quad (3.1)$$

A member of $\text{Succ}_{\mathcal{S}_{\text{hor}}}(H^\partial)$ must preserve the standing quotient $Q_\Sigma(\mathcal{S}_{\text{hor}})$, the exterior witness algebra $\mathcal{A}_{\text{wit}}$, boundary-trace typing, continuation structure, and no-selector/no-repair discipline.

Definition 3.2 (Scope-indexed successor exhaustion). Exterior reconstruction is exhausted at H^∂ , written

$$\text{Exh}_{\mathcal{S}_{\text{hor}}}(H^\partial), \quad (3.2)$$

when

$$\text{Succ}_{\mathcal{S}_{\text{hor}}}(H^\partial) = \emptyset. \quad (3.3)$$

This states only the absence of a standing-preserving admissible successor on the declared exterior scope. It does not state a temporal process, thermodynamic depletion, constraint evolution, ontological disappearance, or dynamical destruction.

Theorem 3.3 (Exterior reconstruction non-guarantee principle). *The admissibility constraints of $\mathcal{S}_{\text{hor}}$ are necessary conditions for lawful exterior reconstruction, but they do not guarantee that $\text{Succ}_{\mathcal{S}_{\text{hor}}}(H^\partial)$ is nonempty.*

Proof. Admissibility constraints specify which continuations would preserve standing, witnesses, boundary traces, and redescription covariance. A specification of lawful continuation is not a generator of a lawful continuation. If every candidate successor fails one of the declared constraints, then the successor set is empty while the constraints remain intact. Treating the constraints as guaranteeing a successor would convert admissibility into a realization principle, which would create standing where no standing-preserving successor exists. That is prohibited by non-generativity and no-repair discipline. $\square$

Lemma 3.4 (Exhaustion is not a horizon process). *If $\text{Succ}_{\mathcal{S}_{\text{hor}}}(H^\partial) = \emptyset$, the correct structural reading is absence of admissible exterior successor, not evaporation, information loss, heat death, causal decay, or temporal exhaustion.*

Proof. The successor set is defined relative to fixed constraints and declared exterior witnesses. Empty successor structure means no candidate object satisfies those constraints. No parameter in the definition records elapsed time, entropy production, energy loss, or changing rules. Therefore exhaustion is realizability failure under fixed conditions, not a physical process. $\square$

Theorem 3.5 (Reconstruction-dependence theorem). *The main horizon theorem is a theorem about same-scope exterior reconstruction, not an ontological theorem about the nonexistence of inaccessible interiors. If $\text{Exh}_{\mathcal{S}_{\text{hor}}}(H^\partial)$ holds, then the admissible classification is exterior-reconstruction exhaustion. It is not licensed to redescribe that exhaustion as a continuing same-scope standing-preserving permanent horizon unless a successor object, certificate, leakage channel, or non-sink quotient is supplied.*

Proof. The declared target is the preservation of exterior standing under admissible continuation. If no successor object exists, then that target has no realization on the fixed scope. This does not imply that every larger mathematical or physical representation lacks interior structure; it implies only that the declared exterior reconstruction problem has no standing-preserving successor. Calling the empty-successor case a continuing admissible permanent horizon would add a successor without witness support, which is repair or inadmissible structure import. Thus the theorem is reconstructive rather than ontological. $\square$

Corollary 3.6 (Boundary persistence without reconstruction persistence). *The exterior admissibility boundary and its rules may remain fixed while no exterior admissible successor exists. Constraint persistence does not entail reconstruction persistence.*

Proof. A fixed rule system may have no solution. Likewise, a fixed exterior admissibility envelope may admit no standing-preserving successor object. The failure is in realizability, not in the rule boundary. $\square$

Remark 3.7 (Use of the beginning/end interior papers). The beginning/end admissible-interior papers are used only for this non-guarantee and successor-exhaustion discipline. Tensor stress, L -metric boundary structure, and capstone α - Ω enumeration are not imported as additional horizon dynamics.

4 Orthodox black-hole interface

This section fixes the ordinary physics interface without attempting to rederive it. In asymptotically flat classical general relativity, the future event horizon is commonly represented as

$$\mathcal{H}^+ = \partial J^-(\mathcal{I}^+), \quad (4.1)$$

the boundary of the causal past of future null infinity. A permanent classical event horizon is therefore globally defined: signals from the black-hole interior do not reach future null infinity. Semiclassical gravity adds Hawking radiation and black-hole thermodynamics, suggesting that black holes radiate with a temperature determined by surface gravity and possess entropy proportional to horizon area. Quantum-information discussions then ask how exterior information recovery can be reconciled with horizon formation, evaporation, and unitarity [3, 4, 6].

The present argument uses none of the detailed numerical apparatus. It only uses the following interface facts as scope data:

- (i) there is an exterior domain in which records and witnesses may be described;
- (ii) there is a candidate horizon boundary that may sever direct exterior access;
- (iii) exterior reconstruction may require standing-bearing reference preservation;
- (iv) a model may or may not supply leakage, radiation, holographic encoding, complementarity, or other anchor-transport data.

Remark 4.1 (Event horizon versus admissible horizon reconstruction). A classical event horizon is a global causal object. An admissible horizon reconstruction is a standing-preserving exterior description object. The theorem targets the second. A classical solution can be a useful idealization while still failing as a terminal standing-preserving same-scope reconstruction if it leaves exterior standing without admissible image.

Definition 4.2 (Permanent horizon relative to exterior scope). Given a partial horizon reconstruction object H^∂ , a candidate boundary B_H is permanent relative to $\mathcal{S}_{\text{hor}}$ when a threatened standing-bearing exterior class is transported into, or becomes dependent on, the permanently isolated envelope and no admissible exterior successor for that class exists:

$$\text{Succ}_{\mathcal{S}_{\text{hor}}}^r(H^\partial) = \emptyset \quad (4.2)$$

for the relevant $r \in Q_\Sigma(\mathcal{S}_{\text{hor}})$, while the model nevertheless presents the situation as a continuing same-scope horizon reconstruction rather than as exterior-successor exhaustion. If only the empty-successor condition is asserted, without a continuing-reconstruction claim, the correct classification is $\text{Exh}_{\mathcal{S}_{\text{hor}}}(H^\partial)$, not an admissible permanent horizon.

Definition 4.3 (Leaky or unstable horizon relative to exterior scope). A candidate boundary B_H is leaky or unstable relative to $\mathcal{S}_{\text{hor}}$ when its evolution supplies exterior witnesses, radiation, boundary correlations, horizon dissolution, or other certified channels sufficient to prevent permanent standing loss. The term “leakage” is structural: it may be realized by Hawking radiation, holographic boundary reconstruction, correlations in exterior fields, or any other admissible mechanism.

5 Standing, admissibility, and horizon reference

Definition 5.1 (Conditioned admissible description). A conditioned admissible description $\hat{D}$ is a description whose reference-bearing content has been restricted to structures anchorable within the

declared scope and stable under the permitted redescription class. Conditioning removes content that is not witnessable, not anchorable, or not licensed by the scope.

Definition 5.2 (Standing). Let $\hat{D}$ be a conditioned admissible description. A reference token r in $\hat{D}$ has standing when:

- (i) r is anchored to a witnessable structure within the admissibility envelope of $\hat{D}$;
- (ii) r remains well-defined under all permitted redescriptions of $\hat{D}$;
- (iii) the referent of r is preserved within scope across admissible continuation.

The standing content of $\hat{D}$ is denoted $\Sigma(\hat{D})$. Standing is not identified with energy, entropy, Shannon information, or Hilbert-space norm. It is a structural measure of reference stability under admissible continuation.

Definition 5.3 (Formal exterior standing profiles). Fix a horizon scope $\mathcal{S}_{\text{hor}}$. Each exterior reference representative $r \in \mathcal{R}_{\text{ext}}$ determines four same-scope profiles:

$$\Phi_W(r) := \{w \in \mathcal{A}_{\text{wit}} : w \text{ re-fixes or distinguishes } r\}, \quad (5.1)$$

$$\Phi_C(r) := \{r' \in \mathcal{R}_{\text{ext}} : r \text{ Cont } r' \text{ by an admissible standing-preserving continuation}\}, \quad (5.2)$$

$$\Phi_B(r) := \text{the boundary-accessibility, leakage, or horizon-trace profile of } r, \quad (5.3)$$

$$\Phi_A(r) := \text{the admissibility, role-type, and transport-status profile of } r. \quad (5.4)$$

Here Φ_A includes whether r is anchor, tensor, or skin content; whether r lies in the domain of T_H ; and whether a certificate or non-sink quotient preserves it.

Definition 5.4 (Exterior standing equivalence and quotient). For $r, r' \in \mathcal{R}_{\text{ext}}$, define

$$r \sim_{\text{ext}} r' \quad (5.5)$$

iff

$$\Phi_W(r) = \Phi_W(r'), \quad \Phi_C(r) = \Phi_C(r'), \quad \Phi_B(r) = \Phi_B(r'), \quad \Phi_A(r) = \Phi_A(r'). \quad (5.6)$$

The exterior standing quotient is

$$\mathcal{Q}_{\text{ext}} := \mathcal{R}_{\text{ext}} / \sim_{\text{ext}}. \quad (5.7)$$

A same-scope reconstruction preserves exterior standing precisely when it preserves $\mathcal{Q}_{\text{ext}}$ and the induced witness, continuation, boundary-accessibility, and admissibility-status profiles.

Proposition 5.5 (Exterior standing equivalence is an equivalence relation). *The relation $\sim_{\text{ext}}$ is reflexive, symmetric, and transitive. Hence $\mathcal{Q}_{\text{ext}}$ is well-defined.*

Proof. Each component profile is compared by equality inside the declared scope. Equality is reflexive, symmetric, and transitive componentwise. The conjunction of equivalence relations is an equivalence relation, so the quotient exists. $\square$

Proposition 5.6 (Standing preservation is quotient preservation). *Let Ψ be a same-scope horizon reconstruction map. Then Ψ preserves exterior standing iff it induces a well-defined map*

$$\tilde{\Psi} : \mathcal{Q}_{\text{ext}} \longrightarrow \mathcal{Q}'_{\text{ext}} \quad (5.8)$$

preserving the four induced profiles $\Phi_W, \Phi_C, \Phi_B, \Phi_A$.

Proof. If Ψ preserves standing, representatives indistinguishable by exterior witnesses, continuation, boundary accessibility, and admissibility status remain indistinguishable after applying Ψ ; otherwise Ψ would introduce a new standing distinction not fixed by the old quotient. Therefore the induced map on classes is well-defined and profile-preserving. Conversely, if the quotient map preserves the four profiles, every standing-bearing reference class is re-fixed through the same admissible exterior data after redescription. That is standing preservation on the fixed scope. $\square$

Theorem 5.7 (Exterior standing quotient uniqueness). *Let $(Y, q, \Theta_W, \Theta_C, \Theta_B, \Theta_A)$ be any admissible presentation of exterior standing content, where $q : \mathcal{R}_{\text{ext}} \rightarrow Y$ is surjective and the four Θ -profiles on Y recover exactly $\Phi_W, \Phi_C, \Phi_B, \Phi_A$. If*

$$q(r) = q(r') \iff r \sim_{\text{ext}} r', \quad (5.9)$$

then there is a unique bijection

$$\chi : \mathcal{Q}_{\text{ext}} \longrightarrow Y, \quad \chi([r]) = q(r). \quad (5.10)$$

Consequently every admissible same-scope presentation of exterior standing factors uniquely through $\mathcal{Q}_{\text{ext}}$ up to witness-preserving bookkeeping equivalence.

Proof. The map $\chi([r]) = q(r)$ is well-defined because q is constant exactly on $\sim_{\text{ext}}$ -classes. It is injective by the same condition and surjective because q is surjective. Uniqueness follows from quotient universality: any profile-preserving map out of $\mathcal{R}_{\text{ext}}$ constant on standing-equivalence classes must factor through the quotient. Thus representative choices, chart choices, hidden labels, and notation changes have no independent standing force unless they alter one of the four quotient profiles, in which case the scope or standing content has changed. $\square$

Theorem 5.8 (Standing conservation). *Under admissible conditioned evolution, standing is conserved:*

$$\Sigma(\hat{D}_{t+\Delta t}) = \Sigma(\hat{D}_t). \quad (5.11)$$

Equivalently, admissible evolution admits neither standing sinks nor standing sources.

Proof. Conditioning restricts descriptions to anchorable content. Admissible evolution transports anchor structures forward, possibly redistributing them across records, fields, correlations, boundary traces, or quotient structures. Redescription invariance prevents standing from being destroyed by a change of representation. Conversely, admissibility prevents standing from being created without derivation from prior anchors. Therefore standing is preserved exactly on the declared scope. $\square$

Definition 5.9 (Reconstruction-theoretic permanence). A horizon boundary B_H is permanent on $\mathcal{S}_{\text{hor}}$ when the standing-transport obstruction for at least one threatened exterior standing class remains non-vanishing throughout every admissible continuation admitted by Cont, unless a leakage channel, standing-transport certificate, or non-sink quotient is explicitly added to the scope. This definition is reconstruction-theoretic rather than coordinate-theoretic. Eternal classical event horizons, long-lived trapping horizons, cosmological horizons, and asymptotic boundaries are not automatically in the same comparison class; each must be typed by its declared exterior witnesses and continuation relation.

Definition 5.10 (Standing sink). A boundary B is a standing sink relative to scope S if there exists a nonempty family T of standing-bearing tokens such that, once transported across B , no admissible future description restricted to S contains a redescription-stable image of T .

Corollary 5.11 (No standing sinks). *No standing sink can arise in admissible conditioned evolution.*

Proof. If a nonempty standing-bearing family T has no admissible future image, then standing strictly decreases on the declared scope. This contradicts standing conservation. $\square$

Remark 5.12 (Standing is not ordinary information). The theorem does not say that Shannon information, von Neumann entropy, or Hilbert-space norm is conserved by itself. Those quantities may be relevant physical witnesses. The theorem concerns whether standing-bearing reference remains admissibly recoverable or quotient-preserved on the fixed exterior scope.

6 Kernel instantiation for exterior horizon reconstruction

The AASC vocabulary used below is local and fixed-domain. It does not add dynamics. It records the minimal constraints already instantiated when a physical description is treated as a coherent standing-bearing reconstruction.

Definition 6.1 (AASC kernel). The AASC kernel is

$$K_{\text{AASC}} = \{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\}, \quad (6.1)$$

where Adm distinguishes licensed from unlicensed same-scope continuation, Stand records reusable reference status, Ref fixes what is being continued, and Irr forbids later repair of a failed or lost identity transport as the same history.

Theorem 6.2 (Kernel instantiation for exterior reconstruction). *Let $CC(\mathcal{S}_{\text{hor}})$ assert that an exterior black-hole history is being evaluated as one coherent same-scope construction of standing-bearing exterior reference. Then K_{AASC} is instantiated on $\mathcal{S}_{\text{hor}}$.*

Proof. If Ref is absent, the exterior reference class whose persistence is being evaluated is not determinate. If Stand is absent, later records, correlations, or boundary images cannot count as reuse of the same physical content. If Adm is absent, there is no distinction between licensed continuation and arbitrary redescription. If Irr is absent, a failed standing transport could be repaired after loss while being counted as the same history, eliminating the difference between preservation and rescue. Each absence destroys the target object rather than yielding a weaker reconstruction. Therefore the kernel is instantiated by the horizon reconstruction problem itself. $\square$

Corollary 6.3 (AASC as proof infrastructure). *The AASC framework is not a competing black-hole dynamics. It supplies the fixed-domain discipline required to decide whether a horizon model preserves standing-bearing exterior reference once the physical envelope has already been declared.*

What kernel denial loses in the exterior horizon theorem

The no-totalization theorem uses all four kernel roles. They are not ornamental assumptions placed on top of black-hole physics. They are the minimal conditions under which the exterior horizon reconstruction target can be stated.

Without Adm , a proposed recovery channel cannot be separated from arbitrary redescription, selector import, carrier laundering, or post-hoc repair. The theorem would then have no way to distinguish an exterior witness from a hidden label, a remnant index, a final-state selector, a

Page-curve slogan, a coordinate extension, or a late holographic dictionary. This is why conservative examples such as ADM/Bondi/Komar mass ledgers, charge as boundary-fixed gauge response, and horizon trace versus horizon label are better entry examples than holography: they already require an admissibility gate before any advanced reconstruction proposal appears.

Without Stand, radiation data, boundary correlations, remnant states, ringdown records, or global states may exist but cannot count as reuse of the same physical content. The distinction between preserving only an exterior parameter quotient and preserving finer infalling reference classes would disappear.

Without Ref, the threatened class in $\mathcal{R}_{\text{ext}}$, the burden being subtracted, and the target of Cert_H are not determinate. A certificate cannot re-fix a referent if no referent has been fixed.

Without Irr, an undefined old-scope value of T_H may be repaired after horizon isolation and then counted as prior preservation. In that case the difference between a standing-transport certificate and an after- the-fact totalization is lost.

Proposition 6.4 (Kernel deletion destroys the exterior horizon target). *Deleting any component of $\mathcal{K}_{\text{AASC}}$ destroys the exterior horizon reconstruction target rather than weakening the theorem. Without Adm, no rule distinguishes a certificate from arbitrary redescription. Without Stand, no later witness can count as reuse of the same content. Without Ref, no threatened exterior class is determinate. Without Irr, failed standing transport can be repaired after loss and counted as preservation. Thus the horizon obstruction is not an optional overlay on black-hole physics; it is the fixed-domain consequence of evaluating one coherent standing-bearing exterior reconstruction problem.*

7 Horizon boundary traces and role-anchor fixation

AASC boundary-trace fixation is the correct import for horizon language. A horizon boundary does not become a reference-preserving object by being named a horizon. It contributes to admissible exterior reconstruction only through stable declared traces in the exterior scope.

Definition 7.1 (Horizon boundary trace). A horizon boundary trace for $\mathcal{S}_{\text{hor}}$ is a declared exterior-accessible predicate or functional

$$\text{Tr}_{\partial H} : X_H \rightarrow \{0, 1\} \quad \text{or} \quad \text{Tr}_{\partial H} : X_H \rightarrow \mathcal{V}_{\partial H} \quad (7.1)$$

whose value is stable under the permitted redescrptions of $\mathcal{S}_{\text{hor}}$ and whose domain is fixed by exterior or boundary witnesses. It is a role anchor only where it is defined and stable. Absence, multiplicity, instability, or post-selected assignment fail closed.

Lemma 7.2 (Boundary trace lemma for horizons). *A candidate horizon boundary may enter a standing-preserving exterior reconstruction only as a typed declared boundary trace. Direct appeal to an inaccessible interior, unspecified global completion, or later-selected dictionary is not itself an admissible horizon trace.*

Proof. A role anchor must fix reference inside the declared domain. An inaccessible interior state, unspecified global object, or post-selected dictionary does not by itself provide a stable exterior predicate. It can become admissibility-bearing only if represented as a declared boundary trace or exterior witness functional. Otherwise the move imports data from outside the scope or selects a representative not licensed by the boundary. $\square$

Theorem 7.3 (Horizon boundary-trace fixation). *Let a standing-bearing exterior reference role be threatened by horizon formation. If internal exterior structure and ordinary redescription leave multiple possible continuations, then the only same-scope asymmetry capable of fixing the role is a unique stable declared boundary trace or certificate. If no such trace exists, fixation fails closed rather than partially succeeding.*

Proof. Reference fixation requires an admissible asymmetry that distinguishes the continued referent from alternatives. If exterior/internal data do not determine a unique continuation, a selector would have to come from the boundary or from outside the declared scope. The AMetric boundary blocks external rankings, probabilities, typicality rules, preferred representatives, and final-state choices unless they are already represented as declared witness data. Therefore the only same-scope role-fixing asymmetry is a stable boundary trace or certificate. If it is absent, nonunique, or unstable, assigning a continuation would require selector import or post-hoc repair; fixation therefore fails closed. $\square$

Corollary 7.4 (Horizon labels are not standing transport). *Calling a boundary an event horizon, apparent horizon, trapping horizon, stretched horizon, or holographic screen does not by itself preserve exterior standing. The label is skin unless it induces a declared boundary trace, witness, certificate, or non-sink quotient on the exterior scope.*

8 Standing-transport certificates

The central admissibility question is whether the horizon model supplies a lawful exterior image of standing-bearing content. The following definition makes the certificate burden explicit.

Definition 8.1 (Standing-transport certificate). Let $T \subseteq \mathcal{R}_{\text{ext}}$ be a family of standing-bearing exterior reference classes whose continuation approaches or crosses a candidate horizon boundary B_H . A standing-transport certificate Cert_H for T is a family of same-scope maps or witness assignments

$$C_t : T \longrightarrow \mathcal{W}_{\text{ext}}(t) \quad (8.1)$$

for sufficiently late exterior stages t , satisfying:

- (C1) **Anchor re-fixation:** $C_t(r)$ contains an anchor sufficient to re-fix the original referent of r within $\mathcal{S}_{\text{hor}}$.
- (C2) **Redescription stability:** the assignment is invariant under all permitted exterior redescrptions.
- (C3) **Continuation covariance:** if $r \text{ Cont } r'$ in the admissible history, then $C_t(r)$ and $C_{t'}(r')$ are related by the declared exterior continuation relation.
- (C4) **Exterior availability:** the witness lies in the admitted exterior witness family $\mathcal{W}_{\text{ext}}$, not solely in an inaccessible interior or external meta-description.
- (C5) **No hidden selector:** the assignment does not require a preferred representative, global index, final-state selector, canonical pairing, or typicality rule not fixed by $\mathcal{S}_{\text{hor}}$.
- (C6) **No post-hoc repair:** the assignment does not create after the fact a transport relation that was undefined on the old carrier.

Definition 8.2 (Certified and uncertified horizon). A horizon reconstruction is certified when every standing-bearing exterior reference class threatened by horizon isolation has a standing-transport certificate. It is uncertified when at least one such class lacks a certificate.

Proposition 8.3 (Certificate sufficiency). *If a horizon model supplies a standing-transport certificate for every threatened standing-bearing exterior reference class, then the horizon is not a standing sink on the declared scope.*

Proof. For each threatened token, the certificate supplies a future exterior witness that re-fixes the referent, preserves redescription stability, and respects continuation. Hence no standing-bearing token disappears from the admissible exterior reconstruction. Therefore the boundary is not a standing sink. $\square$

Proposition 8.4 (Certificate necessity). *If a permanent horizon is not to be a standing sink on a scope carrying a standing-bearing exterior burden, then the model must supply a standing-transport certificate, leakage channel, or non-sink quotient structure.*

Proof. A non-sink boundary must preserve standing-bearing content. Preservation can occur by explicit exterior recovery, leakage, quotient identification, or boundary encoding. Each such route is precisely a certificate or a non-sink quotient. If none exists, some standing-bearing class lacks future exterior image, and the horizon is a standing sink. $\square$

Remark 8.5 (Holography and complementarity). Holography and complementarity are not rejected. They become admissible precisely when they supply the certificate: a redescription-stable exterior or boundary correspondence that preserves standing-bearing reference. A bare assertion that information is “encoded somewhere” is insufficient unless it induces such a same-scope certificate.

9 The horizon reconstruction category

This section fixes the mathematical comparison class. The category is intentionally scope-locked. It formalizes exterior standing-preserving horizon reconstruction, not a complete quantum-gravity model.

Definition 9.1 (Finite horizon reconstruction object). Let $\mathcal{C}_{\text{HOR}}$ be the category whose objects are finite exterior horizon reconstruction systems

$$H = (\mathcal{S}_{\text{hor}}, X_H, B_H, \mathcal{R}_{\text{ext}}, \mathcal{W}_{\text{ext}}, \text{Cont}, T_H, \text{Cert}_H, \Delta_\Sigma, \text{Env}), \quad (9.1)$$

where $\mathcal{S}_{\text{hor}}$ is the fixed exterior scope, X_H is the source-horizon-exterior carrier, B_H is the candidate horizon boundary, $\mathcal{R}_{\text{ext}}$ is the standing-bearing exterior reference family, $\mathcal{W}_{\text{ext}}$ is the exterior witness family, Cont is admissible continuation, T_H is the standing-transport relation or partial operator, Cert_H is the certificate family, Δ_Σ records standing surplus/loss status, and Env records physical envelope data not derived by the theorem.

Definition 9.2 (Admissible horizon reconstruction morphism). A morphism $\Phi : H \rightarrow H'$ in $\mathcal{C}_{\text{HOR}}$ is an admissible reconstruction-preserving map satisfying:

M1. Exterior-anchor preservation. Φ sends standing-bearing exterior reference classes to standing-bearing exterior reference classes without changing the declared burden.

- M2. Witness preservation.** Φ preserves the exterior witness family up to lawful redescription.
- M3. Boundary typing.** Φ preserves whether B_H is exterior boundary, horizon boundary, leakage boundary, or scope-changing interface.
- M4. Continuation preservation.** If $r \text{ Cont } r'$ is admissible, then $\Phi(r) \text{ Cont } \Phi(r')$ is admissible; equivalences preserve the converse.
- M5. Transport-status preservation.** Φ preserves whether a standing-transport certificate exists. It may not turn an uncertified transport failure into a certified recovery without adding admitted data or changing scope.
- M6. Standing ledger preservation.** Φ preserves Δ_Σ : it may not erase a nonzero standing loss while preserving the same exterior burden.
- M7. Skin covariance.** Coordinate charts, slicings, labels, and representative choices may vary only when exterior anchors, witnesses, continuation, and certificate status are preserved.
- M8. No-selector/no-repair discipline.** Φ may not introduce a hidden global index, final-state selector, preferred representative, post-hoc relabeling, or repaired transport relation not fixed by $\mathcal{S}_{\text{hor}}$.

Definition 9.3 (Admissible horizon equivalence). Two horizon reconstruction objects $H, H' \in \text{Obj}(\mathcal{C}_{\text{HOR}})$ are admissibly equivalent, written $H \sim_{\text{adm}} H'$, iff there are admissible reconstruction morphisms both ways preserving exterior anchors, witnesses, continuation, transport status, and standing ledger. Uniqueness below means uniqueness of $[H]_{\text{adm}}$, not uniqueness of notation, foliation, verbal horizon label, or representative chart.

Proposition 9.4 ($\mathcal{C}_{\text{HOR}}$ is a category). *Finite horizon reconstruction objects with admissible horizon reconstruction morphisms form a category.*

Proof. The identity morphism preserves every component of a reconstruction object: scope, boundary type, references, witnesses, continuation, transport status, certificate family, standing ledger, and envelope. If $\Phi : H \rightarrow H'$ and $\Psi : H' \rightarrow H''$ are admissible, then $\Psi \circ \Phi$ preserves anchors, witnesses, boundary typing, continuation, transport status, standing ledger, and skin covariance by transitivity. A hidden selector or repair cannot appear in the composite unless it appears in one of the factors, which admissibility forbids. Associativity is inherited from ordinary map composition. $\square$

Definition 9.5 (Exterior witness category and reconstruction functor). Let $\mathcal{W}_{\text{HOR}}$ be the category whose objects are exterior witness packages

$$W_H = (\mathcal{R}_{\text{ext}}, \mathcal{W}_{\text{ext}}, \text{Cont}, \text{Cert}_H, \Delta_\Sigma), \quad (9.2)$$

with morphisms given by witness-preserving redescrptions. Define the reconstruction functor

$$F_H : \mathcal{C}_{\text{HOR}} \longrightarrow \mathcal{W}_{\text{HOR}} \quad (9.3)$$

by sending a horizon reconstruction object to its exterior witness package and an admissible morphism to the induced witness-preserving map.

Proposition 9.6 (Functoriality of exterior reconstruction). *The assignment F_H is a functor. It sends identities to identities and composition of admissible reconstruction morphisms to composition of witness-preserving maps.*

Proof. The identity on a reconstruction object induces the identity on its witness package. If Φ and Ψ are composable morphisms, the induced map on references, witnesses, continuation, certificate status, and standing ledger is the componentwise composition. Thus $F_H(\Psi \circ \Phi) = F_H(\Psi) \circ F_H(\Phi)$. $\square$

Definition 9.7 (Universal admissible horizon quotient). Let

$$\pi_H : \mathcal{C}_{\text{HOR}} \longrightarrow \mathcal{Q}_{\text{HOR}} \quad (9.4)$$

be the quotient functor identifying exactly those reconstruction objects and morphisms related by admissible equivalence. The category $\mathcal{Q}_{\text{HOR}}$ is the universal admissible horizon reconstruction quotient of the declared exterior scope.

Theorem 9.8 (Universal factorization through $\mathcal{Q}_{\text{HOR}}$). *For any category $\mathcal{D}$ and any functor $G : \mathcal{C}_{\text{HOR}} \rightarrow \mathcal{D}$ that is constant on admissible-equivalence classes of objects and morphisms, there exists a unique functor $\overline{G} : \mathcal{Q}_{\text{HOR}} \rightarrow \mathcal{D}$ such that*

$$G = \overline{G} \circ \pi_H. \quad (9.5)$$

Consequently every same-scope admissible horizon reconstruction factors through $\mathcal{Q}_{\text{HOR}}$ up to witness-preserving bookkeeping equivalence.

Proof. Define $\overline{G}([H]_{\text{adm}}) = G(H)$ and define $\overline{G}$ on morphism classes by applying G to any representative. This is well-defined because G is constant on admissible-equivalence classes. The factorization identity follows by construction. Uniqueness follows because π_H is surjective onto quotient object and morphism classes. $\square$

Theorem 9.9 (Same-scope closure theorem). *A horizon reconstruction proposal remains same-scope iff it preserves*

$$(\mathcal{A}_{\text{ext}}, \mathcal{A}_{\partial H}, \mathcal{A}_{\text{wit}}, \mathcal{R}_{\text{ext}}, \mathcal{Q}_{\text{ext}}, T_H, \text{Cont}, \text{Succ}_{\mathcal{S}_{\text{hor}}}, \text{Env}) \quad (9.6)$$

up to $\mathcal{Q}_{\text{ext}}$ -equivalence and skin-level redescription. If it preserves these data, it represents the same object of $\mathcal{Q}_{\text{HOR}}^{\partial}$ up to bookkeeping equivalence. If it changes any one of them in a standing-bearing way, it changes the witness package, boundary typing, continuation structure, operator domain, successor structure, or physical scope.

Proof. The displayed tuple lists the data needed to evaluate the same exterior standing-reconstruction problem: exterior and boundary witnesses, standing quotient, partial standing transport, admissible continuation, successor availability, and physical envelope. Preservation of the tuple preserves the object classified by the theorem; any remaining changes are representative choices and hence skin. A standing-bearing difference must enter by changing a profile of $\mathcal{Q}_{\text{ext}}$, the domain of T_H , the successor set, or the envelope. Each change is either a new witness/certificate, a repair or selector, or a scope change. No further same-scope locus is available by the anchor/tensor/skin decomposition and fixed-domain exhaustion. $\square$

10 Witness completion for exterior horizon scope

A common objection says that the standing is preserved by a hidden witness sector not included in the exterior description. The correct AASC response is not to deny larger-domain physics. It is to ask whether the alleged witness is same-scope: does it factor through the declared exterior witness algebra, change the witness package, remain disconnected, or enlarge the physical scope?

Definition 10.1 (Exterior witness algebra). The exterior witness algebra $\mathcal{A}_{\text{wit}}$ is the closure of the exterior records, boundary traces, asymptotic observables, radiation data, correlation functionals, and admissible reconstruction maps declared in $\mathcal{W}_{\text{ext}}$. It is minimal relative to $\mathcal{S}_{\text{hor}}$ when deleting any generator removes an admitted exterior standing witness. It is admissibly complete relative to $\mathcal{S}_{\text{hor}}$ when every same-scope standing-reconstruction witness factors through $\mathcal{A}_{\text{wit}}$, Cont, and any declared certificate data.

Definition 10.2 (Operational exterior witness). An exterior witness $w \in \mathcal{W}_{\text{ext}}$ is operationally admitted only if it contributes to at least one of the following within the declared exterior problem: boundary trace fixation, asymptotic or finite-region readout, exterior correlation comparison, radiation/field functional comparison, admissible reconstruction of a standing quotient, or certificate verification. A symbol that does none of these is not an exterior witness in $\mathcal{S}_{\text{hor}}$; it is skin, disconnected larger-domain data, or a hidden selector depending on how it is used.

Lemma 10.3 (No arbitrary witness import). *A proposed additional witness Q cannot become standing-bearing merely by being named physically important, computationally useful, or globally available. It is same-scope only when it has an operational exterior coupling to $\mathcal{A}_{\text{wit}}$ or supplies a certificate-preserving map into $\mathcal{R}_{\text{ext}}/\sim_{\text{stand}}$.*

Proof. The declared exterior problem is fixed by its admitted readouts and reconstruction functionals. If Q has no coupling to those readouts and no certificate map, it does not re-fix an exterior standing class. Treating it as authoritative despite that absence introduces a selector or carrier not fixed by the exterior domain. If Q changes the readouts or admissible reconstruction target, the scope has been enlarged. $\square$

Theorem 10.4 (Exterior witness accessibility theorem). *A candidate Q belongs to the admissible exterior witness algebra iff it is reconstructively accessible from the declared exterior continuation structure without hidden selector import, post-hoc repair, causal or asymptotic scope enlargement, new standing-generating structure, or representative-only distinction. Equivalently, every proposed exterior witness falls into exactly one of the following cases:*

- (i) *it is generated by $\mathcal{A}_{\text{wit}}$, Cont, Leak, Cert_H , or the declared non-sink quotient;*
- (ii) *it changes Φ_W for some class of $\mathcal{Q}_{\text{ext}}$ and therefore changes the witness profile;*
- (iii) *it is disconnected from exterior reconstruction and has no standing force on $\mathcal{S}_{\text{hor}}$;*
- (iv) *it imports a selector, carrier, or admissibility-generating source not fixed by $\mathcal{Q}_{\text{ext}}$;*
- (v) *it changes physical or causal scope.*

Proof. A witness is admissible only by doing reconstructive work on the declared exterior problem. If it is accessible through the declared exterior continuation and reconstruction maps, it is generated by the old witness algebra and associated certificates. If it changes which standing classes are distinguished or re-fixed, it changes the witness profile. If it has no exterior coupling, it is disconnected from the reconstruction task. If it is used to choose or authorize a reconstruction without exterior coupling, it is selector or carrier import. If it requires new causal access, asymptotic data, or a new sector, it changes scope. These cases exhaust the ways a symbol can acquire witness status. $\square$

Definition 10.5 (Witness-complete exterior horizon scope). A horizon scope is witness-complete for exterior standing when every admissible readout of threatened standing-bearing reference is

generated by $\mathcal{A}_{\text{wit}}$ together with declared leakage, certificate, or quotient structure. This is a scope theorem, not a metaphysical completeness claim about all possible black-hole physics.

Theorem 10.6 (Minimal exterior witness completion). *Within a witness-complete exterior horizon scope, $\mathcal{A}_{\text{wit}}$ is the minimal admissibly complete witness algebra for the declared exterior reconstruction problem. Any proposed additional horizon-relevant witness Q falls into exactly one of four classes:*

- (i) Q is generated by $\mathcal{A}_{\text{wit}}$, Cont, Leak, Cert_H , or the declared non-sink quotient;
- (ii) Q changes exterior reconstruction functionals, detector access, boundary traces, or standing statistics;
- (iii) Q is empirically or admissibly disconnected from exterior standing reconstruction;
- (iv) Q extends the physical scope beyond the declared horizon reconstruction problem.

Thus no hidden witness sector supplies an uncertified same-scope permanent-horizon rescue.

Proof. Completeness is relative to the declared exterior problem: admitted exterior reconstruction proceeds through the witnesses, traces, continuation maps, leakage channels, and certificate data already included in $\mathcal{A}_{\text{wit}}$. If Q is available without changing that reconstruction problem, then it is definable from those data and belongs to class (i). If it alters what the exterior can measure, reconstruct, or compare, it changes the witness object and is class (ii). If it has no admissible exterior coupling, it is not a witness for exterior standing and is class (iii). If it adds a new boundary algebra, asymptotic region, detector channel, interior branch, quantum-gravity sector, or external code not factoring through $\mathcal{A}_{\text{wit}}$, it changes scope and is class (iv). These alternatives exhaust the same-scope status of an alleged witness. $\square$

Corollary 10.7 (Hidden witness sector collapse). *A hidden horizon sector preserving all old exterior witnesses is either definitional bookkeeping or an inadmissible selector. If it changes exterior standing, it changes scope or supplies the required certificate.*

Proposition 10.8 (Exterior witness completion blocks pure interior storage). *A purely interior storage claim preserves exterior standing only if it induces a witness in $\mathcal{A}_{\text{wit}}$ or a certificate Cert_H . Otherwise it is either disconnected from the exterior reconstruction object or belongs to a larger-domain theory.*

Proof. Exterior standing is re-fixability under admitted exterior continuation. Pure interior storage supplies no such re-fixation unless it induces a map into the declared exterior witnesses. That induced map is precisely a witness or certificate. Without it, the storage claim may be meaningful in another scope but not as same-scope exterior reconstruction. $\square$

11 Operator-domain obstruction: standing transport as a partial operator

The obstruction form treats an uncertified permanent horizon as an attempted totalization of a partial standing-transport operator.

Definition 11.1 (Horizon carrier decomposition). Let

$$X_H = E_{\text{ext}} \cup E_{\partial H} \cup E_{\text{int}} \tag{11.1}$$

be the source-horizon-exterior carrier of a declared horizon reconstruction object. Here E_{ext} is the exterior witness envelope, $E_{\partial H}$ is the boundary/horizon trace envelope, and E_{int} is the interior or permanently isolated envelope relative to the exterior scope.

Definition 11.2 (Partial standing-transport operator). The horizon standing-transport assignment is a partial operator

$$T_H : X_H \rightharpoonup \mathcal{R}_{\text{ext}} \quad (11.2)$$

with

$$\text{dom}(T_H) = E_{\text{ext}} \cup E_{\partial H}^{\text{cert}}, \quad (11.3)$$

where $E_{\partial H}^{\text{cert}}$ consists of boundary or leakage data carrying a standing-transport certificate. On an uncertified permanent interior region, $E_{\text{int}} \cap \text{dom}(T_H) = \emptyset$.

Remark 11.3 (Undefined is not ignorance). Undefinedness of T_H on E_{int} is not a third standing value and not temporary ignorance. It is non-membership in the admissible operator domain. The interior may exist in a larger mathematical or physical description; what is absent is an exterior standing-transport assignment on the declared scope.

Definition 11.4 (Partial horizon reconstruction category). The partial horizon reconstruction category $\mathcal{C}_{\text{HOR}}^{\partial}$ refines $\mathcal{C}_{\text{HOR}}$ by making the domain of T_H intrinsic to each object. An object is a tuple

$$H^{\partial} = (H, T_H, E_{\text{ext}}, E_{\partial H}, E_{\text{int}}). \quad (11.4)$$

A morphism $\Phi^{\partial} : H^{\partial} \rightarrow H'^{\partial}$ is an admissible morphism $\Phi : H \rightarrow H'$ in $\mathcal{C}_{\text{HOR}}$ preserving partial domains:

$$\Phi(E_{\text{ext}}) \subseteq E'_{\text{ext}}, \quad \Phi(E_{\partial H}) \subseteq E'_{\partial H}, \quad \Phi(\text{dom}(T_H)) \subseteq \text{dom}(T'_H). \quad (11.5)$$

No morphism in $\mathcal{C}_{\text{HOR}}^{\partial}$ may send an old uncertified interior point into the defined domain of standing transport unless the physical scope is enlarged or a certificate is supplied inside the old scope.

Proposition 11.5 ($\mathcal{C}_{\text{HOR}}^{\partial}$ is a category). *Partial horizon reconstruction objects and domain-preserving partial morphisms form a category.*

Proof. Identity maps preserve all envelopes and partial domains. Compositions preserve inclusions by transitivity. The underlying maps compose in $\mathcal{C}_{\text{HOR}}$, and associativity is inherited from ordinary composition. $\square$

Theorem 11.6 (Horizon transport totalization obstruction). *No admissible same-scope extension totalizes T_H on an uncertified permanent interior while preserving the original exterior witnesses. Equivalently, there is no same-scope operator*

$$T_H^* : X_H \longrightarrow \mathcal{R}_{\text{ext}} \quad (11.6)$$

that extends T_H , is defined on E_{int} , and preserves the exterior witness package, unless a standing-transport certificate exists or the scope changes.

Proof. By definition, T_H is undefined on an uncertified permanent interior relative to the exterior scope. A totalized operator T_H^* would make an old undefined standing-transport assignment defined on the old carrier while agreeing with T_H on the old exterior domain. In AASC partial-domain closure, admissible extensions preserve old operator domains on the original carrier; making an undefined old-scope application defined is repair, generation, selector import, or scope change. If a certificate exists, the relevant data were already in $E_{\partial H}^{\text{cert}}$ and the horizon is not the uncertified case. If no certificate exists, totalization is inadmissible. $\square$

Definition 11.7 (Internal horizon totalization obstruction). For a partial horizon reconstruction object H^∂ , define the internal horizon totalization obstruction

$$O_H(H^\partial) \quad (11.7)$$

as the internal obstruction, in $\mathcal{C}_{\text{HOR}}^\partial$, to the attempted totalization

$$T_H : E_{\text{ext}} \cup E_{\partial H}^{\text{cert}} \rightarrow \mathcal{R}_{\text{ext}} \quad \rightsquigarrow \quad T_H^* : X_H \rightarrow \mathcal{R}_{\text{ext}}. \quad (11.8)$$

By definition, $O_H(H^\partial) = 0$ only if there exists an admissible same-scope totalization preserving exterior anchors, witnesses, continuation, and standing ledger. Otherwise $O_H(H^\partial) \neq 0$.

Remark 11.8 (Status of O_H). The symbol O_H is not asserted to be a standard cohomological, K -theoretic, or AQFT invariant. It records the internal admissible-totalization obstruction in the declared horizon reconstruction category.

Theorem 11.9 (Nonvanishing of the horizon obstruction). *For every witness-complete exterior horizon reconstruction object with standing-bearing exterior burden and an uncertified permanent horizon,*

$$O_H(H^\partial) \neq 0. \quad (11.9)$$

Proof. If $O_H(H^\partial) = 0$, then an admissible same-scope totalization T_H^* exists. That contradicts the horizon transport totalization obstruction. The alleged vanishing witness must either supply a certificate, import hidden data, repair the old domain, introduce a selector, or change physical scope. The first exits the uncertified case; the remaining routes are inadmissible or larger-scope. Hence the obstruction is nonzero. $\square$

Definition 11.10 (Horizon operator-obstruction classes). Any attempted vanishing witness for O_H must instantiate at least one of the following classes:

- E1. Selector insertion.** A hidden global index, preferred representative, final-state condition, typicality rule, canonical pairing, or dictionary chooses the missing exterior reconstruction.
- E2. Repair or generator.** An undefined old-scope standing-transport assignment is made defined after horizon isolation.
- E3. Carrier laundering.** A derived item–entropy, probability, symmetry, code word, dual reconstruction claim, remnant label, or explanatory virtue—is treated as primitive authorization for standing transport.
- E4. Scope transport.** The exterior witness algebra, asymptotic boundary, detector access, interior branch, quantum-gravity sector, or reconstruction target is changed while presented as the same exterior problem.
- E5. Bookkeeping duplication.** A hidden symbol preserves every old exterior witness and adds no standing-bearing distinction.

Theorem 11.11 (Operator-obstruction classification). *Every same-scope attempt to totalize standing transport across an uncertified permanent horizon falls into E1–E5. Classes E1–E4 are inadmissible or larger-scope; E5 is inert bookkeeping. Hence no additional same-scope operator rescue remains.*

Proof. A totalization attempt must acquire reconstruction force in one of five ways. It may select a continuation not fixed by old witnesses; then E1 holds. It may make an old undefined operator application defined; then E2 holds. It may authorize the assignment by a derived object not itself an exterior witness; then E3 holds. It may change the domain on which reconstruction is evaluated; then E4 holds. Or it may add a symbol that changes no standing-bearing exterior fact; then E5 holds. Anchor/tensor/skin role decomposition and fixed-domain exhaustion leave no further admissibility-bearing locus. E1 violates no-selector discipline, E2 violates irreversibility and partial-domain closure, E3 imports an unlicensed carrier, E4 changes the theorem object, and E5 is skin. $\square$

Definition 11.12 (Universal partial horizon quotient). Let

$$\pi_H^\partial : \mathcal{C}_{\text{HOR}}^\partial \longrightarrow \mathcal{Q}_{\text{HOR}}^\partial \quad (11.10)$$

be the quotient functor identifying partial horizon reconstruction objects exactly when they are related by admissible domain-preserving morphisms both ways and preserve exterior witnesses. The quotient $\mathcal{Q}_{\text{HOR}}^\partial$ is universal for admissible partial horizon reconstruction on the declared scope.

Theorem 11.13 (Universal quotient form of horizon obstruction). *In $\mathcal{Q}_{\text{HOR}}^\partial$, an uncertified permanent-horizon object with standing-bearing exterior burden has no admissible totalized representative. Any alleged second representative either maps to the same quotient object, supplies a certificate, violates partial-domain preservation, or changes physical scope.*

Proof. The quotient identifies exactly witness-preserving admissible redescrptions. A totalized representative would define T_H on the old uncertified interior while preserving the old exterior witnesses, violating domain preservation. If the representative avoids this violation, it must either leave the same quotient object unchanged, add certificate data, or alter scope. These are exhaustive by the definition of $\mathcal{Q}_{\text{HOR}}^\partial$. $\square$

12 Reconstruction realizability and successor exhaustion

The operator obstruction says an uncertified totalization is not admissible. The admissible-interior closure discipline adds the complementary point: the absence of a lawful successor is allowed as exhaustion, but exhaustion may not be relabeled as a successful same-scope reconstruction.

Definition 12.1 (Realized exterior continuation). A realized exterior continuation of H^∂ is an element of $\text{Succ}_{\mathcal{S}_{\text{hor}}}(H^\partial)$ that preserves at least one threatened standing class by leakage, certificate, quotient identification, or declared exterior witness transport.

Definition 12.2 (False permanence). A false permanence claim is a claim that B_H remains a continuing same-scope standing-preserving permanent horizon when $\text{Succ}_{\mathcal{S}_{\text{hor}}}(H^\partial) = \emptyset$ and no certificate, leakage channel, or non-sink quotient is supplied.

Theorem 12.3 (Reconstruction realizability exhaustion). *For a witness-complete horizon object with threatened exterior standing, exactly one of the following same-scope classifications is available:*

- (i) *a realized admissible exterior continuation exists through leakage, certificate, or non-sink quotient;*
- (ii) *$\text{Succ}_{\mathcal{S}_{\text{hor}}}(H^\partial) = \emptyset$ and the exterior reconstruction has exhausted;*

(iii) a proposed continuation uses selector import, repair, carrier laundering, inadmissible structure import, or scope transport.

Cases (i) and (ii) are admissible classifications. Case (iii) is not an admissible same-scope continuation.

Proof. If a standing-preserving successor exists, it is case (i). If none exists, the successor set is empty and case (ii) holds by definition. Any alleged continuation in the empty-successor case must introduce definedness, standing, witness access, or authority not supplied by the declared scope. By operator-domain closure and admissibility-generating structure exhaustion, such acquisition can only occur by selector, repair, carrier laundering, authority import, or scope change, which is case (iii). These alternatives are exhaustive because a continuation either belongs to the successor set or does not. $\square$

Corollary 12.4 (No persistence-by-admissibility). *Admissibility does not force a black-hole exterior reconstruction to persist. It only determines whether a proposed successor, if offered, is lawful.*

Proof. Immediate from the exterior reconstruction non-guarantee principle and the realizability exhaustion theorem. $\square$

Corollary 12.5 (No false permanence). *A false permanence claim is inadmissible. If exterior successor structure is absent, the allowable classification is exhaustion, not standing-preserving permanent horizon continuation.*

Proof. False permanence asserts a continuing same-scope reconstruction while the successor set is empty and no certificate or non-sink quotient exists. That requires adding a successor not in $\text{Succ}_{\mathcal{S}_{\text{hor}}}(H^\partial)$, hence repair, selector import, inadmissible structure import, or scope transport. $\square$

13 No-permanent-horizon reconstruction theorem

We can now state the hardened theorem.

Definition 13.1 (Uncertified permanent standing sink). An uncertified permanent horizon B_H is an uncertified permanent standing sink when there exists a nonempty family $T \subseteq \mathcal{R}_{\text{ext}}$ such that members of T are transported into or rendered dependent on E_{int} , no member has a future admissible exterior image in $\mathcal{W}_{\text{ext}}$, and no standing-transport certificate or non-sink quotient exists.

Theorem 13.2 (No-permanent-horizon reconstruction theorem). *Let $H^\partial \in \text{Obj}(\mathcal{C}_{\text{HOR}}^\partial)$ be a witness-complete exterior horizon reconstruction object. Suppose:*

- (i) $\mathcal{S}_{\text{hor}}$ carries a nonempty standing-bearing exterior burden;
- (ii) B_H is permanent relative to $\mathcal{S}_{\text{hor}}$ in the successor-set sense;
- (iii) no standing-transport certificate, leakage channel, or non-sink quotient exists for at least one threatened reference class;
- (iv) the model is nevertheless claimed to be a continuing same-scope standing-preserving horizon reconstruction rather than an exhausted exterior reconstruction.

Then H^∂ is not an admissible same-scope permanent-horizon reconstruction object.

Proof. By (i), at least one exterior reference class requires standing-preserving continuation if the declared reconstruction is to continue. By (ii) and (iii), the successor set for at least one threatened standing class is empty: no exterior witness, certificate, leakage channel, or quotient supplies a future admissible image. The admissible-interior non-guarantee principle permits this to be classified as exterior-successor exhaustion. Premise (iv), however, refuses that classification and presents the empty-successor situation as continuing standing-preserving permanence. That move either treats the horizon as a standing sink, contradicting no-standing-sinks, or totalizes T_H on the old uncertified carrier, contradicting the horizon transport totalization obstruction. Hence the object is inadmissible as a same-scope permanent-horizon reconstruction. $\square$

Corollary 13.3 (Leakage, evaporation, or certificate necessity). *Any admissible continuing horizon model on a standing-bearing exterior scope must instantiate at least one of the following:*

$$\text{leakage} \quad \vee \quad \text{evaporation} \quad \vee \quad \text{certified exterior reconstruction} \quad \vee \quad \text{non-sink quotient structure.} \quad (13.1)$$

If none is available, the admissible classification is exterior-successor exhaustion, not continuing permanent-horizon reconstruction.

Proof. If none occurs and the model is still presented as continuing reconstruction, the horizon is an uncertified permanent standing sink, excluded by the no reconstruction-generating structure theorem. Without the continuing-reconstruction claim, the case is classified as successor exhaustion. $\square$

Corollary 13.4 (Evaporation as minimal structural resolution). *When no stronger certified boundary reconstruction or non-sink quotient is supplied, horizon instability or evaporation is the minimal admissible resolution of the standing-sink obstruction.*

Proof. The obstruction is removed by preventing permanent standing loss. If no holographic, complementarity, remnant, or quotient certificate is supplied, the remaining direct way to avoid permanent standing loss is leakage or horizon dissolution. Complete evaporation is one minimal structural realization of that alternative; the theorem does not compute the dynamical channel. $\square$

14 Finite toy collapse witness and transport failure

The previous sections are abstract by design. This section gives a small finite witness showing exactly where the obstruction occurs.

Definition 14.1 (Finite toy horizon reconstruction object). A finite toy horizon reconstruction object is

$$\mathsf{T}_H = (R, W, B, \ell, c, \tau), \quad (14.1)$$

where $R = \{r_1, r_2, \dots, r_n\}$ is a finite family of standing-bearing exterior reference classes, $W = \{w_1, w_2, \dots, w_m\}$ is a finite exterior witness set, B is a candidate horizon boundary, $\ell \in \{0, 1\}$ is a leakage indicator, $c \in \{0, 1\}$ is a certificate indicator, and $\tau : R \rightarrow W$ is the partial standing-transport assignment.

Definition 14.2 (Toy domain rule). The toy domain rule is:

$$\text{dom}(\tau) = \begin{cases} R, & \ell = 1 \text{ or } c = 1, \\ R \setminus R_{\text{hid}}, & \ell = 0, c = 0, \end{cases} \quad (14.2)$$

where $\emptyset \neq R_{\text{hid}} \subseteq R$ is the set of reference classes rendered dependent on the permanently isolated horizon interior. The case $\ell = 1$ represents leakage or evaporation; the case $c = 1$ represents certified boundary/holographic/remnant reconstruction; the case $\ell = c = 0$ is the uncertified permanent-horizon case.

Case	Domain of τ	Status
$\ell = 1, c = 0$	all of R	leakage supplies late witnesses; no standing sink.
$\ell = 0, c = 1$	all of R	a certificate supplies reconstruction; no standing sink.
$\ell = 0, c = 0$	$R \setminus R_{\text{hid}}$	at least one standing-bearing class has no exterior image; permanent standing sink.

Example 14.3 (Three toy cases).

Theorem 14.4 (Toy totalization failure). *In the uncertified case $\ell = c = 0$, no same-scope extension may replace τ by a total function $\tau^* : R \rightarrow W$ while preserving the old witness set W , old boundary typing, and old continuation relation.*

Proof. For each $r \in R_{\text{hid}}$, $\tau(r)$ is undefined by the toy domain rule. A totalization τ^* chooses a witness $w \in W$ for r despite the absence of leakage or certificate. If w is determined by old witnesses, then $\tau(r)$ was not undefined. If w is not so determined, the choice is a hidden selector. If a new witness is added, W changes. If the assignment is made later after loss, it is repair. Thus same-scope totalization fails. $\square$

Example 14.5 (Explicit finite partial-algebra obstruction witness). Let

$$X = \{e_1, e_2, b, h_1, h_2\}, \quad \mathcal{R}_{\text{ext}} = \{r_1, r_2, r_b\}, \quad \mathcal{W}_{\text{ext}} = \{w_1, w_2, w_b\}. \quad (14.3)$$

The exterior points e_i carry admitted witnesses w_i , the boundary point b carries w_b , and h_i are permanent-horizon hidden points. Define the partial transport operator

$$T_H(e_i) = r_i, \quad T_H(b) = r_b, \quad T_H(h_i) \uparrow, \quad (14.4)$$

where $\uparrow$ denotes non-membership in the operator domain. Let $\mathcal{A}_{\text{wit}} = \text{Alg}\{w_1, w_2, w_b\}$ and assume no leakage or certificate. An attempted totalization assigns

$$T_H^*(h_i) = r_{\sigma(i)} \quad (14.5)$$

for some choice $\sigma : \{1, 2\} \rightarrow \{1, 2, b\}$.

Theorem 14.6 (Explicit finite partial-algebra obstruction). *In the finite partial-algebra witness above, no same-scope totalization T_H^* is admissible. Hence $O_H \neq 0$ for the toy uncertified permanent horizon.*

Proof. If σ is fixed by $\mathcal{A}_{\text{wit}}$, then the hidden point h_i already had an exterior witness image and $T_H(h_i)$ was not undefined. If σ is not fixed by $\mathcal{A}_{\text{wit}}$, the assignment chooses a representative by hidden selector. If a remnant index, dual dictionary, Page-curve inference, or global state is used to fix σ , that object either factors through $\mathcal{A}_{\text{wit}}$ as a certificate or is carrier laundering/scope transport. If the assignment is added only after the undefinedness is recognized, it is repair or generation of an old partial operation. If no assignment is made, the successor set for the threatened class is empty. These are all acquisition-exhaustion cases, so no same-scope totalization survives. $\square$

Proposition 14.7 (Toy horizon failure certificate). *In the toy witness, standing transport can be harmless before horizon isolation but cannot be continued into the late exterior stage as exact standing-preserving identity unless the late exterior scope contains leakage or a standing-transport certificate.*

Proof. Before isolation, every standing-bearing reference may still be associated with an exterior or boundary witness. After isolation with $\ell = c = 0$, the reference classes in R_{hid} lack exterior images. The late exterior burden still requires them to stand. The only admissible ways to satisfy that burden are leakage, certificate, or quotient identification; each makes τ defined for the threatened class. Without one of these, the assignment is undefined and totalization is forbidden. $\square$

Lemma 14.8 (No global-reconstruction repair). *Let H^∂ contain an uncertified permanent horizon relative to $\mathcal{S}_{\text{hor}}$. A later global description cannot retroactively restore exterior standing as the same admissible exterior history unless it induces a standing-transport certificate on $\mathcal{S}_{\text{hor}}$.*

Proof. A global description may contain mathematical data outside the declared exterior scope. For that data to restore exterior standing, it must provide an admissible exterior witness image preserving anchor identity, redescription stability, and continuation. That is the certificate. If no certificate is induced, the global description is either disconnected from exterior standing, a hidden selector, a post-hoc repair, or a changed scope. $\square$

15 Boundary-to-boundary horizon reconstruction

The admissible reconstruction object is not necessarily direct recovery of an interior classical state. It is any boundary-to-boundary or exterior-to-exterior structure preserving standing-bearing reference.

Definition 15.1 (Boundary-to-boundary standing kernel). A boundary-to-boundary standing kernel is a map or family of functionals

$$K_\Sigma : \mathcal{R}_{\text{ext}}(t_{\text{in}}) \times \mathcal{W}_{\text{ext}}(t_{\text{out}}) \longrightarrow \mathcal{V}_\Sigma \quad (15.1)$$

that composes an initial standing-bearing exterior reference, the declared horizon evolution, and a future exterior witness into a redescription-stable reconstruction value. $\mathcal{V}_\Sigma$ is any value object sufficient for standing re-fixation: record class, correlation class, boundary functional, radiation code, or quotient witness.

Theorem 15.2 (Boundary-to-boundary reconstruction theorem). *On a witness-complete exterior horizon scope, a horizon model preserves standing only through a boundary-to-boundary standing kernel, leakage channel, or quotient-equivalent certificate preserving exterior anchors, witnesses, continuation, and no-repair discipline.*

Proof. The initial reference supplies the anchor. The horizon evolution supplies the transport or obstruction. The late exterior witness supplies the re-fixation target. Their admissible composition is a boundary-to-boundary standing kernel or an equivalent certificate. If no such composition exists, the threatened reference class has no admissible future exterior image and standing is lost. That is a standing sink. $\square$

Corollary 15.3 (Hawking radiation as candidate dynamical realization). *Hawking radiation, or any leakage channel carrying redescription-stable exterior witnesses sufficient to re-fix standing-bearing content, is a candidate dynamical realization of the structural requirement. The theorem does not derive the radiation spectrum or rate.*

16 Global retention versus exterior reconstruction

Global retention is a common objection. It says that whatever is lost to exterior access still exists in the global state, the full spacetime, the exact wavefunction, the dual theory, or the quantum-gravity completion. The theorem does not deny such global retention. It denies that global retention alone is same-scope exterior standing preservation.

Definition 16.1 (Global retention claim). A global retention claim asserts that information, norm, microstate data, correlations, or reference-relevant structure persist in a larger total description G even though they are not available as standing-bearing witnesses in $\mathcal{A}_{\text{wit}}$.

Definition 16.2 (Exterior recoverability certificate). A global retention claim induces an exterior recoverability certificate when there exists a same-scope map

$$\text{Rec}_G : G|_{\mathcal{S}_{\text{hor}}} \longrightarrow \mathcal{A}_{\text{wit}} \quad (16.1)$$

that re-fixes every threatened $r \in \mathcal{R}_{\text{ext}}$, preserves redescription stability, respects Cont, and introduces no hidden selector or post-hoc repair. In that case Rec_G is a standing-transport certificate.

Theorem 16.3 (Global retention is not exterior reconstruction). *A globally existing correlation structure preserves exterior standing only if it induces an exterior recoverability certificate on $\mathcal{S}_{\text{hor}}$. If no such certificate is induced, the global retention claim is hidden global indexing, larger-domain data, or bookkeeping without exterior standing force.*

Proof. Exterior standing requires re-fixability by admitted exterior witnesses. A total or global description can contain data not present in the declared exterior scope. For those data to preserve standing on $\mathcal{S}_{\text{hor}}$, they must determine, from the admissible exterior witness package, a redescription-stable image of each threatened reference class. This is exactly the recoverability certificate. If the map is absent, the data do not function as exterior standing. Treating them as decisive then either imports a hidden global index, changes the reconstruction domain, or leaves all old exterior witnesses unchanged and is bookkeeping. $\square$

Corollary 16.4 (Unitarity separation). *A unitary total evolution, by itself, is not a standing-transport certificate. It becomes sufficient for the present theorem only when it induces the exterior recoverability certificate required by $\mathcal{S}_{\text{hor}}$.*

Corollary 16.5 (Holographic separation). *A holographic dual, by itself, is not a same-scope exterior certificate. It becomes sufficient precisely when the dual encoding induces a redescription-stable anchor map into the declared exterior witness algebra.*

Theorem 16.6 (Dual reconstruction scope theorem). *A dual, holographic, or error-correcting representation preserves same-scope exterior admissibility only if its reconstruction map preserves $\mathcal{Q}_{\text{ext}}$, $\mathcal{A}_{\text{wit}}$, and Cont without introducing a second standing-generating structure. If it does preserve*

them, it is a standing-transport certificate or a bookkeeping-equivalent presentation of the same reconstruction sector. If it does not, it instantiates selector insertion, repair/generation, carrier laundering, witness modification, or scope transport.

Proof. A dual representation can matter for the old exterior scope only through the standing quotient, witnesses, and continuation relation. Preservation of those data gives either a certificate, when threatened classes are re-fixed, or an equivalent representation of the same quotient object. Failure to preserve them means the dual map either chooses a representative not fixed by old witnesses, makes an old undefined transport defined, treats the dual dictionary as primitive authority, modifies the witness profile, or changes the domain. Those alternatives are precisely the acquisition-exhaustion classes below. $\square$

Corollary 16.7 (Asymptotic recovery separation). *A claim of infinite-time or asymptotic recovery is same-scope only if the declared continuation relation admits the relevant future witness as part of $\mathcal{A}_{\text{wit}}$. Otherwise it is unavailable repair or a different reconstruction scope.*

17 Remnant compression obstruction

Stable remnants deserve a separate closure because they are the most compact way to retain a permanent horizon-like endpoint while denying information loss. The AASC issue is not whether a remnant can exist in a larger theory. The issue is whether remnant storage preserves exterior standing without an exterior reconstruction map.

Definition 17.1 (Remnant reconstruction claim). A remnant reconstruction claim asserts that a late stable object R_{rem} stores the standing-bearing content threatened by horizon isolation while the exterior scope remains $\mathcal{S}_{\text{hor}}$.

Definition 17.2 (Admissible remnant map). An admissible remnant map is a same-scope map

$$\rho_{\text{rem}} : R_{\text{rem}} \times \mathcal{A}_{\text{wit}} \longrightarrow \mathcal{R}_{\text{ext}} \quad (17.1)$$

that re-fixes the threatened exterior reference classes, preserves redescription stability, respects Cont, and does not depend on a hidden index, unbounded unexplained compression selector, or post-hoc pairing rule.

Theorem 17.3 (Remnant compression obstruction). *A stable remnant preserves exterior standing on $\mathcal{S}_{\text{hor}}$ only if it exposes an admissible remnant map to the exterior witness algebra or is quotient-equivalent to a certified non-sink reconstruction. If no such map exists, the remnant is a standing sink, hidden compression index, carrier-laundering device, or larger-scope storage claim.*

Proof. If the remnant exposes an admissible map ρ_{rem} , then threatened standing is re-fixed through exterior witnesses and the remnant is not an uncertified sink. If it exposes no such map, then the exterior scope cannot distinguish which threatened reference classes are preserved by the remnant. Treating the remnant as preserving them requires a hidden index, a compression selector, or a post-hoc pairing between old references and remnant states. Those moves violate no-selector/no-repair discipline. If the remnant's authority comes from a larger theory not factoring through $\mathcal{A}_{\text{wit}}$, the scope has changed. Therefore an uncertified remnant does not rescue a permanent horizon on the old exterior scope. $\square$

Corollary 17.4 (Planck remnant storage is not enough). *An arbitrarily large or Planck-scale remnant Hilbert space is insufficient for this theorem unless it supplies the exterior remnant map. Capacity is not standing; exterior re-fixability is the relevant certificate.*

18 Conservative reconstruction collapse for hidden horizon sectors

Theorem 18.1 (Conservative hidden-sector collapse). *Let H^* be an admissibility-preserving extension of a horizon reconstruction object that adds a hidden interior, branch, index, remnant label, or horizon microstate symbol while preserving all old exterior witnesses, standing references, continuation relations, and certificate status. Then H^* is definitional/bookkeeping-equivalent to the original reconstruction on the old scope. If it is not bookkeeping-equivalent, it changes witnesses, introduces a selector, repairs an undefined standing assignment, imports a carrier, or changes scope.*

Proof. Conservativity on the old scope means every old standing-bearing witness is unchanged. A new symbol that changes no old witness is skin unless explicitly definable from the old witness package. If it is not definable but treated as standing-bearing, its authority comes from a hidden selector, repaired totalization, carrier laundering, or scope transport. Those routes are inadmissible or larger-domain moves. Hence conservative hidden sectors collapse to bookkeeping. $\square$

Corollary 18.2 (Interior-only standing does not preserve exterior standing). *A standing-bearing assignment confined to an inaccessible permanent interior does not preserve exterior standing unless it induces a certificate on the exterior scope.*

19 No admissibility-generating structure for exterior reconstruction

The preceding sections classify hidden sectors and remnant compression. One further objection must be treated explicitly: the claim that some authority—a duality, a global unitary description, an expert reconstruction procedure, a Page-curve calculation, a remnant bookkeeping index, a final-state rule, or a quantum-gravity completion—can by itself authorize exterior standing recovery. The AASC authority papers supply the required closure: authority may organize, compute, endorse, or describe a reconstruction, but it cannot generate admissibility or standing on the old scope [25, 26].

Definition 19.1 (Reconstruction-generating structure). A reconstruction-generating structure (authority, in the older interpretive terminology) for $\mathcal{S}_{\text{hor}}$ is any structure $\mathcal{U}$ claimed to generate, restore, override, certify, or substitute for admissible exterior standing reconstruction across B_H . Examples include holographic dictionaries, global-state claims, Page-curve inferences, remnant indices, asymptotic completions, final-state maps, interior selectors, baby-universe labels, institutional declarations, expert endorsements, and computational reconstruction algorithms.

Definition 19.2 (Reconstruction-generating structure relevance). A reconstruction-generating structure $\mathcal{U}$ is reconstruction-irrelevant on $\mathcal{S}_{\text{hor}}$ when it preserves every exterior witness profile, standing quotient, boundary trace, continuation relation, and certificate status. It is reconstruction-relevant when it changes any admissibility status, transport definedness, standing quotient, certificate status, or exterior reconstruction availability.

Theorem 19.3 (No reconstruction-generating structure theorem). *No reconstruction-generating structure can generate, restore, override, or substitute for admissible exterior standing reconstruction on a fixed horizon scope. If $\mathcal{U}$ is reconstruction-irrelevant, it is bookkeeping-equivalent on the old exterior scope. If $\mathcal{U}$ is reconstruction-relevant, it necessarily performs selector import, repair or generation of an undefined old-scope transport assignment, carrier laundering, or scope transport.*

Proof. Assume first that $\mathcal{U}$ preserves all exterior witnesses, standing quotients, boundary traces, continuation relations, and certificate status. Then it changes no admissibility-bearing fact on $\mathcal{S}_{\text{hor}}$; by skin covariance and conservative reconstruction collapse, it is bookkeeping-equivalent on the old scope. It may be useful notation or a larger explanatory presentation, but it has not generated exterior reconstruction.

Assume instead that $\mathcal{U}$ changes an admissibility-bearing fact. If it changes transport definedness across an uncertified permanent horizon, it totalizes T_H and violates partial-domain closure. If it fixes a representative not selected by $\mathcal{A}_{\text{wit}}$, it imports a selector. If it treats a derived quantity, computational success, expert assertion, dual reconstruction claim, global norm, or remnant label as primitive standing authorization, it launders a carrier. If it adds new readouts, sectors, boundaries, or continuation rules, it changes scope. These are exactly the inadmissible or larger-domain acquisition modes. No third case remains. $\square$

Corollary 19.4 (Reconstruction performance is not standing). *High-fidelity reconstruction, successful simulation, algorithmic recovery, Page-curve agreement, or dual-theory calculability does not by itself create exterior standing. Such performance is admissibility-relevant only when it induces a certificate-preserving map into the declared exterior witness algebra; otherwise it is operationally useful but reconstruction-irrelevant.*

Proof. Performance concerns success relative to a task or representation. Exterior standing concerns reference re-fixation inside $\mathcal{S}_{\text{hor}}$. A procedure that does not induce a map into $\mathcal{A}_{\text{wit}}$ or $\mathcal{R}_{\text{ext}}/\sim_{\text{stand}}$ changes no standing-bearing exterior fact. A procedure that does induce such a map supplies the required certificate. A procedure asserted to change standing without such a map is reconstruction-generating structure import, excluded by the no reconstruction-generating structure theorem. $\square$

Corollary 19.5 (Holographic and global reconstruction separation). *A holographic dual, global unitary description, or quantum-gravity completion preserves exterior standing on $\mathcal{S}_{\text{hor}}$ only if it induces a lawful same-scope certificate on $\mathcal{A}_{\text{wit}}$ and $\mathcal{R}_{\text{ext}}/\sim_{\text{stand}}$. Otherwise it is either a bookkeeping-equivalent reformulation, a larger-domain theory, or an inadmissible reconstruction-authority claim.*

Proof. The dual or global description is either witness-preserving on the old exterior scope, witness-changing, or certificate-inducing. Witness-preserving structure is bookkeeping on $\mathcal{S}_{\text{hor}}$. Witness-changing structure changes scope. Certificate-inducing structure satisfies the theorem's required alternative and no longer represents an uncertified permanent horizon. $\square$

Theorem 19.6 (Exterior admissibility-generating structure exhaustion). *Every same-scope admissibility-bearing reconstruction-generating structure for horizon continuation arises from one of exactly three admissible loci: the declared exterior witness algebra $\mathcal{A}_{\text{wit}}$, a lawful boundary/leakage/certificate transport map, or the admissible continuation relation Cont . Any alleged fourth admissibility-generating structure collapses into bookkeeping, selector import, repair/generation, carrier laundering, or scope change.*

Proof. A standing-bearing exterior reconstruction must re-fix a reference class. Re-fixation can occur through admitted exterior witnesses, through a certified boundary/leakage transport map, or through the continuation relation that preserves the standing quotient. If a proposed structure does not operate through any of these, it either changes no exterior standing fact and is bookkeeping, chooses a representative not fixed by the domain, repairs an undefined transport assignment, treats a derived object as primitive standing authorization, or changes the domain. Anchor/tensor/skin role typing and fixed-domain exhaustion exclude any further governance-bearing locus. $\square$

20 Same-scope reconstruction acquisition exhaustion

Before listing black-hole-specific rescue routes, we state the final meta-exhaustion theorem. It replaces a merely enumerative route table with a proof that every same-scope acquisition of new exterior reconstruction force has one of the listed forms.

Theorem 20.1 (Reconstruction acquisition exhaustion theorem). *Every same-scope attempt to acquire new standing-bearing exterior reconstruction force has a unique route image, up to overlap by bookkeeping notation, in one of the following acquisition modes:*

- (A1) **Witness modification:** *it changes Φ_W or the generated algebra $\mathcal{A}_{\text{wit}}$.*
- (A2) **Boundary retyping:** *it changes Φ_B , the horizon boundary role, or the admitted accessibility profile.*
- (A3) **Continuation modification:** *it changes Φ_C , Cont , or the exterior successor structure.*
- (A4) **Selector insertion:** *it chooses a representative or continuation not fixed by $\mathcal{Q}_{\text{ext}}$.*
- (A5) **Repair/generation:** *it makes an old undefined standing-transport operation defined on the old carrier.*
- (A6) **Carrier laundering:** *it treats global state, duality, probability, entropy, performance, remnant index, computation, or explanatory usefulness as primitive standing authorization.*
- (A7) **Scope transport:** *it changes the declared reconstruction domain, causal access, asymptotic boundary, detector/witness algebra, or physical envelope.*
- (A8) **Successor exhaustion:** *it accepts $\text{Succ}_{\text{S}_{\text{hor}}}^T(H) = \emptyset$ and therefore makes no same-scope preservation claim.*
- (A9) **Bookkeeping equivalence:** *it changes none of $\Phi_W, \Phi_C, \Phi_B, \Phi_A, T_H, \text{Succ}_{\text{S}_{\text{hor}}}$, or Env .*

No tenth same-scope acquisition mode remains.

Proof. A standing-bearing reconstruction difference must alter the exterior standing quotient or the ability to continue one of its classes. If it alters the quotient, it changes a witness profile, boundary profile, continuation profile, or admissibility/transport profile. Witness, boundary, and continuation changes are A1–A3. A changed admissibility or transport profile is admissible only when supplied by declared witnesses or certificates; otherwise the acquisition must choose a representative (A4), repair an undefined domain (A5), or treat some derived carrier as primitive authority (A6). If the domain or envelope is changed, the route is A7. If no admissible continuation exists and no preservation claim is made, the route is A8. If none of the profiles, domains, successors, or envelope data change, the difference is A9. Anchor/tensor/skin role typing leaves no fourth descriptive role, partial-domain

closure leaves no way to define an old undefined operation, and fixed-domain exhaustion leaves no further standing-bearing locus. Hence no tenth route remains. $\square$

Corollary 20.2 (Final acquisition exclusion). *A permanent-horizon rescue that neither supplies a certificate nor accepts successor exhaustion must instantiate selector insertion, repair/generation, carrier laundering, witness/boundary/continuation modification, or scope transport. Hence an uncertified permanent horizon cannot become admissible by appeal to an additional reconstruction-generating structure.*

Proof. Immediate from the acquisition-exhaustion theorem after excluding witness-preserving equivalence, certificate-preserving reconstruction, successor exhaustion, and bookkeeping. $\square$

21 Permanent-horizon route exhaustion

This section closes the same-scope escape space.

Definition 21.1 (Permanent-horizon rescue claim). A permanent-horizon rescue claim asserts that an uncertified permanent horizon can remain an admissible same-scope exterior reconstruction object while preserving the declared standing-bearing exterior burden.

Definition 21.2 (Horizon rescue route family). Every rescue claim has a route image in the following family:

Route	Escape claim	Closure response
H1	Global information exists somewhere.	Not sufficient unless it induces same-scope exterior standing recovery. Otherwise larger-domain retention is not exterior reconstruction.
H2	Holographic encoding preserves the interior.	Admissible only with explicit anchor transport/redescription correspondence inducing a certificate on $\mathcal{A}_{\text{wit}}$. Otherwise it is reconstruction-authority laundering, informal assertion, or hidden dictionary.
H3	Complementarity avoids contradiction.	Admissible only if it supplies redescription-stable anchor correspondence for exterior standing. Otherwise it changes observer scope or loses standing.
H4	Stable remnants store the lost content.	A remnant is a compression sink unless it supplies an admissible exterior reconstruction map with finite standing capacity and no hidden selector.
H5	Infinite-time or asymptotic recovery restores standing.	Same-scope only if the declared exterior continuation includes an admissible future witness. Otherwise it is scope transport or unavailable repair.

H6	Observer-relative horizons remove the sink.	Either boundary typing changes by observer scope or the exterior burden remains. It is not a same-scope rescue without certificate.
H7	Interior-only standing preserves the content.	Interior standing is not exterior standing preservation unless it induces an exterior certificate.
H8	Bookkeeping redescription keeps all information.	If it changes no witness it is skin; if it changes standing it imports hidden load or repair.
H9	Final-state/post-selection repair restores information.	A final-state selector is not same-scope admissible unless declared as physical witness data; otherwise it is post-hoc repair.
H10	Baby universe or disconnected branch carries standing.	This changes scope unless the old exterior receives an admissible standing image.
H11	Quantum-gravity completion will resolve it.	A completion is relevant only through the exterior certificate it induces. Without that, it is larger-domain promissory data or authority without standing.
H12	Classical GR event horizons are exact.	Classical exactness is envelope data, not standing preservation. The theorem concerns admissible exterior reconstruction.
H13	A duality, algorithm, expert judgment, or Page-curve calculation authorizes recovery.	Reconstruction-authority does not generate standing. It is relevant only through a same-scope certificate; otherwise it is bookkeeping, carrier laundering, or scope change.
H14	Admissibility constraints guarantee exterior reconstruction persistence.	The non-guarantee principle blocks this: constraints define lawful successors; they do not ensure a nonempty successor set.
H15	Cosmological, Rindler, or observer horizons are treated as the same counterexample.	They require separate scope typing. Same-scope relevance exists only when the same standing-bearing exterior burden is preserved.

Theorem 21.3 (Nonempty route-image theorem). *Every permanent-horizon rescue claim on a fixed exterior scope with standing-bearing burden has nonempty route image in H1–H15.*

Proof. The route table is not an empirical taxonomy of black-hole models; it is a classification of admissibility-acquisition modes available within the declared reconstruction scope.

A rescue claim must say where threatened standing goes or why it need not be recovered. If it appeals to larger global data, it is H1 or H11. If it appeals to boundary encoding, it is H2. If

it appeals to observer-dependent descriptions, it is H3 or H6. If it stores content in an object, it is H4. If recovery is delayed indefinitely, it is H5. If standing remains only inside, it is H7. If the rescue is representational, it is H8. If it uses late constraints or post-selection, it is H9. If it sends content to another branch, it is H10. If it appeals to classical exactness, it is H12. If it appeals to a duality, algorithm, Page-curve inference, expert judgment, or completion as a reconstruction-generating source rather than as a certificate map, it is H13. If it claims admissibility itself guarantees persistence, it is H14. If it invokes cosmological or observer horizons as the same object without preserving the fixed exterior burden, it is H15. These alternatives exhaust the ways to retain a permanent horizon while denying standing loss or denying the standing burden. $\square$

Theorem 21.4 (Route-collapse theorem). *No route H1–H15 survives as an uncertified same-scope permanent-horizon rescue. Each route either supplies a certificate, collapses to bookkeeping, violates no-selector/no-repair discipline, or changes physical scope.*

Proof. H1 supplies no exterior standing unless it induces a certificate. H2 succeeds precisely when it supplies anchor transport; otherwise it is reconstruction-authority laundering, hidden dictionary, or informal assertion. H3 succeeds only with redescription-stable correspondence; otherwise observer scope changes. H4 fails unless remnant compression is certified and standing-preserving. H5 is same-scope only if the recovery witness is admitted by the continuation relation. H6 changes boundary typing unless exterior burden is preserved by certificate. H7 is not exterior preservation. H8 is skin if witness-inert and illicit if standing-bearing. H9 is post-hoc repair unless declared as witness data in the original scope. H10 changes scope unless exterior recovery is induced. H11 is larger-domain physics unless it gives the certificate. H12 is an envelope assertion, not a standing reconstruction. H13 collapses by no admissibility-generating structure: performance, dual calculability, or endorsement does not create standing unless it induces a certificate. H14 collapses by exterior reconstruction non-guarantee: admissibility does not force realizability. H15 is scope transport unless the cosmological or observer-horizon model preserves the same standing-bearing exterior reconstruction burden. Thus each route either satisfies the theorem’s required alternative, classifies the case as exhaustion, or fails as a same-scope rescue. $\square$

Theorem 21.5 (Symmetric route exhaustion for permanent horizons). *On a fixed witness-complete exterior horizon scope requiring standing-bearing reference, an uncertified permanent horizon has no surviving same-scope rescue route.*

Proof. By the nonempty route-image theorem, every rescue claim has a route in H1–H15. By the route-collapse theorem, no route remains as an uncertified same-scope permanent-horizon rescue. Therefore the rescue space is exhausted. $\square$

22 Relation to standard black-hole physics

Hawking evaporation. The theorem does not derive Hawking radiation. It states a structural requirement that a horizon model must avoid permanent standing sinks. Hawking radiation is one candidate dynamical route by which a horizon can become leaky and eventually non-permanent relative to exterior reconstruction. Hawking evaporation, holographic reconstruction, leakage, or non-sink quotient structure are candidate realizations of the structural requirement, not consequences derived at the level of rate, mechanism, or microstate dynamics.

Black-hole information paradox. The theorem is aligned with the information paradox but is not identical to it. The paradox is often framed around unitarity, entropy, and state reconstruction. The present theorem is framed around standing-bearing exterior reference. Global information preservation is not enough unless it induces admissible exterior standing reconstruction.

Reconstruction non-guarantee. The theorem does not assume that the exterior reconstruction problem must always possess a successor. If no standing-preserving successor exists under fixed exterior constraints, the case is classified as exterior-successor exhaustion. What is excluded is the further claim that such exhaustion is nevertheless a continuing standing-preserving permanent horizon.

Global retention. The global-retention theorem is the main separation principle for a common level error. A total wavefunction, full spacetime, exact dual state, or quantum-gravity completion may retain correlations. For this manuscript, those correlations become standing-preserving only when they induce the exterior recoverability certificate on $\mathcal{A}_{\text{wit}}$.

Page curve and unitarity. A Page curve may witness information return in a quantum-information sense. For the present theorem it is sufficient only if the recovered correlations support standing-transport certificates for the declared exterior reference classes.

Holography. Holography is an admissible solution candidate, not an opponent. If the holographic boundary theory gives a redescription-stable anchor correspondence preserving exterior standing, it supplies a certificate. If it merely asserts encoding without anchor transport, it does not close the standing-sink route.

Reconstruction-generating structures. No calculation, dual description, institutional declaration, expert endorsement, or successful algorithm creates standing by reconstruction-authority alone. It becomes relevant to the present theorem only through the certificate it induces on $\mathcal{A}_{\text{wit}}$ and $\mathcal{R}_{\text{ext}}/\sim_{\text{stand}}$. This prevents the common slide from “the information is represented somewhere” to “the old exterior scope has recovered standing.”

Complementarity. Complementarity preserves standing only if it supplies explicit correspondence between exterior and infalling descriptions under admissible redescription. Without such correspondence, it is an observer-scope change rather than same-scope standing preservation.

Stable remnants. Remnants can be admissible only if the remnant state remains an exterior witness-bearing object sufficient to re-fix the threatened standing classes without hidden selectors or unbounded compression sinks. Otherwise remnants are standing reservoirs inaccessible to exterior reconstruction.

Classical horizons. Classical event horizons remain useful idealizations. The theorem says that an exact permanent horizon with no leakage, certificate, or non-sink quotient is not admissible as a standing-preserving exterior reconstruction object.

Cosmological and observer horizons. Cosmological horizons, Rindler horizons, and observer-relative horizons require separate scope typing. The present theorem applies to them only when a fixed standing-bearing exterior reconstruction burden is declared and the alleged horizon functions as a permanent obstruction to that burden. Otherwise they are not same-scope counterexamples.

Quantum gravity. A quantum-gravity completion can satisfy the theorem if it supplies the relevant standing-preserving exterior map. The theorem does not select the completion.

23 AASC dependency ledger

Target step	AASC closure role	Horizon translation
Coherent exterior reconstruction	Kernel non-optionality	Exterior black-hole histories instantiate admissibility, standing, reference, and irreversibility.
Standing preservation	Standing-admissibility identity closure	No same-scope evolution may destroy standing or create replacement standing by repair.
Boundary/horizon trace	Boundary-trace fixation	Horizon and exterior records are anchors only where witness-supported; absent or nonunique traces fail closed.
Exterior witness completion	Constraint architectures for coherent evaluability	Same-scope standing witnesses factor through the declared exterior witness algebra or change scope.
No hidden recovery selector	AMetric boundary	No global index, typicality rule, final-state selector, or canonical representative has authority unless fixed by the declared scope.
Partial transport	Partial-domain closure	Undefined standing transport through an uncertified permanent horizon cannot be totalized on the old carrier.
Admissible-successor exhaustion	End/beginning admissible interior closure	Admissibility constraints define lawful exterior successors but do not guarantee that any successor is realizable.

Global retention separation	Admissible record construction and quantum boundary compatibility	Global retention is not exterior standing unless it induces an exterior recoverability certificate.
Remnant compression	Conservative reconstruction collapse	Stable storage preserves standing only through an admissible exterior remnant map or non-sink quotient.
Reconstruction-authority recovery	Authority impossibility / liability without standing	Duality, performance, endorsement, or declared recovery cannot create standing unless it factors through the exterior witness algebra or certificate map.
Hidden interior sector	Conservative reconstruction collapse	Hidden symbols preserving all exterior witnesses are bookkeeping unless they import new standing-bearing load.
Role decomposition	Anchor/tensor/skin	Exterior anchors, propagation/witness tensor, and representational skin exhaust same-scope roles.
Escape closure	Fixed-domain exhaustion	Every rescue route changes anchor, tensor, witness, continuation, scope, supplies a certificate, classifies exhaustion, or is inert.
Standard physics bridge	Physical-envelope classification	Rates, spectra, entropy, geometry, and quantum-gravity inputs remain envelope data, not derived by the theorem.

24 Objection and response ledger

Attack	Closure response
“This just restates the information paradox.”	It is a narrower reconstruction theorem. It concerns standing-bearing exterior reference, not only entropy, norm, or information.
“Global unitarity already preserves the information.”	Global retention is sufficient only if it induces a same-scope exterior standing-transport certificate.

“Classical GR has event horizons.”	The theorem does not deny classical solutions. It denies uncertified permanent horizons as standing-preserving exterior reconstruction objects.
“Holography solves this.”	Holography is admissible when it supplies explicit anchor transport inducing a certificate on the exterior witness algebra. Without that, it is reconstruction-authority laundering or larger-domain reformulation.
“Complementarity avoids the problem.”	Complementarity must preserve standing under redescription correspondence. Otherwise it changes observer scope.
“Stable remnants can store everything.”	Storage is not reconstruction. A remnant must supply an admissible exterior standing map; capacity alone is not standing.
“A Planck remnant has enough states.”	A large state space is envelope capacity. Without a witness-complete remnant map to $\mathcal{A}_{\text{wit}}$, it is hidden compression or scope change.
“AdS/CFT already proves boundary unitarity.”	A dual proof is relevant here only through the exterior anchor map it induces on the declared witness algebra; otherwise it is larger-domain retention.
“Evaporation is dynamical, not structural.”	The theorem does not derive a mechanism. It proves a structural requirement that some leakage/certificate/non-sink route exist.
“Standing is nonstandard.”	Standing is defined operationally as reference stability under admissible exterior continuation and redescription.
“This assumes quantum mechanics.”	The core obstruction is scope/admissibility based. Quantum assumptions enter only through the envelope if a model declares them.
“This is semantic.”	The theorem forbids a class of reconstruction operators and same-scope totalizations. It is a structural constraint on admissible descriptions.
“A later theory will recover everything.”	A later theory is relevant only through the exterior certificate it supplies. Otherwise it changes scope.
“The AQFT language is overclaimed.”	The appendix is only a boundary-functional embedding, not a full Haag-Kastler or DHR construction.
“A reconstruction algorithm succeeds.”	Performance is not standing. Algorithmic success matters only if it induces a certificate-preserving map into the declared exterior witness algebra.
“A quantum-gravity authority declares recovery.”	Authority cannot generate admissibility. If it changes no exterior witness, it is bookkeeping; if it changes standing without a certificate, it is repair, selector import, or scope change.
“You assume admissible reconstruction must persist.”	No. The non-guarantee principle says admissibility constraints are necessary, not sufficient. Empty successor structure is exhaustion, not a forced continuation.

“You are making an ontological claim about inaccessible interiors.”	No. The reconstruction-dependence theorem classifies same-scope exterior reconstruction. It does not assert that inaccessible interiors do or do not exist in larger domains.
“An exhausted exterior successor set is just another permanent horizon.”	No. Empty successor structure is classified as exterior reconstruction exhaustion. Calling it continuing standing-preserving permanence would require repair or inadmissible structure import.
“What about de Sitter or observer horizons?”	Those require separate scope typing. They are relevant only when a fixed standing-bearing exterior reconstruction burden is declared.
“What exact object is being quotiented?”	The exterior standing object is $\mathcal{Q}_{\text{ext}}(\mathcal{S}_{\text{hor}}) = \mathcal{R}_{\text{ext}} / \sim_{\text{ext}}$, where equivalence preserves witness, continuation, boundary-accessibility, and admissibility-status profiles.
“What would count as a counterexample?”	A counterexample must preserve $\mathcal{Q}_{\text{ext}}$, $\mathcal{A}_{\text{wit}}$, boundary accessibility, continuation, and successor structure while totalizing the old partial transport operator without selector, repair, carrier laundering, scope transport, exhaustion, or bookkeeping.

Counterexample precision. A counterexample must preserve the declared exterior quotient and witness algebra while supplying standing-bearing recovery without changing scope, importing a selector, repairing undefined transport, laundering a carrier, or collapsing into bookkeeping.

Arc handoff summary

The exterior theorem ends with standing break and undefined old-scope transport, not with an interior ontology. What is exported forward is the horizon-isolated carrier as a standing-transport failure site: a class lacks an exterior image, same-scope totalization is blocked, and any rescue must be a certificate, leakage route, non-sink quotient, explicit exhaustion, or declared scope change. The next paper alone asks what interior role-status remains after that break.

25 Conclusion

Series-position conclusion. The result exported by this paper is deliberately narrow: fixed-scope exterior standing break, undefined old-scope standing transport on the threatened horizon-isolated carrier, and the exclusion of uncertified permanent-horizon reconstruction. It does not classify the carrier as residual coherence or as an AMetric endpoint. Those are the next two arc steps.

The compact horizon-instability note proved that permanent horizons are standing sinks unless compensating mechanisms exist, and standing sinks violate standing conservation. This manuscript converts that consequence into a universal obstruction theorem.

The resulting form is: exterior horizon reconstruction instantiates admissibility, standing, reference, and irreversibility; threatened exterior standing requires transport, leakage, certificate, or quotient

preservation; an uncertified permanent horizon makes standing transport partial; totalizing that partial operator on the old carrier is inadmissible; admissibility constraints do not guarantee that a successor exists; empty successor structure is classified as exhaustion rather than persistence; the horizon obstruction O_H is therefore nonzero; every same-scope permanent-horizon rescue route collapses into bookkeeping, selector import, repair, certificate, exhaustion, or scope change.

Thus black-hole evaporation is not derived here as a spectrum, rate, or microscopic mechanism. It is derived as one minimal structural route by which a continuing admissible horizon model can avoid becoming a permanent standing sink. More generally, the theorem forces the following disjunction for any continuing admissible horizon model on a standing-bearing exterior scope:

$$\boxed{\text{leakage} \vee \text{evaporation} \vee \text{certified boundary reconstruction} \vee \text{non-sink quotient.}} \quad (25.1)$$

An exact permanent horizon with none of these is not an admissible same-scope reconstruction object. If the successor set is empty, the admissible statement is exterior-reconstruction exhaustion, not successful permanent-horizon continuation.

Counterexample criterion. A counterexample would require a fixed exterior horizon reconstruction scope with a standing-bearing reconstruction of permanently hidden reference classes that preserves $\mathcal{Q}_{\text{ext}}(\mathcal{S}_{\text{hor}})$, $\mathcal{A}_{\text{wit}}$, boundary accessibility, admissible continuation, and successor structure, while avoiding selector insertion, repair/generation, carrier laundering, scope transport, successor exhaustion, and bookkeeping equivalence. Equivalently, it would exhibit an uncertified totalization of the old partial standing-transport operator that is nevertheless domain-preserving. The reconstruction acquisition exhaustion theorem says that no such mode is available inside the declared scope.

A AQFT-style boundary-functional embedding

This appendix does not construct a full Haag-Kastler model of black-hole evaporation, prove locality, covariance, isotony, or perform DHR superselection classification. It records a localized-algebraic reading of the finite exterior reconstruction theorem.

Let O_{ext} be an exterior region, $O_{\partial H}$ a horizon boundary region, and O_{int} an interior or inaccessible region. Associate schematic algebras

$$\mathcal{A}_{\text{ext}} = \mathcal{A}(O_{\text{ext}}), \quad \mathcal{A}_{\partial H} = \mathcal{A}(O_{\partial H}), \quad \mathcal{A}_{\text{int}} = \mathcal{A}(O_{\text{int}}). \quad (\text{A.1})$$

Exterior standing witnesses are functionals on $\mathcal{A}_{\text{ext}}$ or on boundary-induced subalgebras of $\mathcal{A}_{\partial H}$ that are admitted by the exterior scope. The interior algebra does not by itself preserve exterior standing unless it induces a boundary or exterior functional.

Theorem A.1 (Boundary-functional exhaustion theorem). *Let ρ and ρ' be boundary-functional horizon representations preserving all declared exterior functionals $\omega_{r \rightarrow w}(t)$. Any propagation-local or interior selector distinguishing ρ from ρ' is exactly one of the following: boundary-definable from $\mathcal{A}_{\text{ext}} \vee \mathcal{A}_{\partial H}$, boundary-functionally inert, a hidden selector not fixed by the exterior sector, or a scope-extending addition of new functionals or algebras.*

Proof. If the selector is definable from the exterior/boundary algebra, it is part of the same boundary-functional sector. If it preserves all functionals but is not definable from them, it contributes no

exterior witness distinction and is inert unless treated as standing-bearing. If treated as standing-bearing without boundary definition, it is a hidden selector. If it changes or adds functionals, algebras, or regions, the boundary-functional scope has changed. These four cases exhaust the algebraic status of a selector relative to the declared functionals. $\square$

Theorem A.2 (Boundary-functional selector collapse). *In the boundary-functional reading, a propagation-local or interior standing selector that preserves all exterior boundary functionals is either definable from the exterior boundary algebra and hence bookkeeping-equivalent, or it is not an admissibly witnessed exterior reconstruction-generating structure. If it changes a boundary functional, it changes the reconstruction sector.*

Proof. Exterior reconstruction sectors are individuated by the declared boundary functionals. A selector preserving all of them induces no new exterior standing distinction unless it is treated as authority from outside the boundary algebra. If it is definable from the boundary algebra, it is a redescription of the same sector. If it is not definable but standing-bearing, it is a hidden authority source, blocked by exterior admissibility-generating structure exhaustion. If it changes the functionals, the sector is different. $\square$

Definition A.3 (Boundary-functional horizon representation). A boundary-functional horizon representation is a representation ρ of the displayed exterior-boundary algebraic data on a Hilbert space $\mathcal{K}_\rho$ such that the exterior witness functionals

$$\omega_{r \rightarrow w}^\rho(t) = \omega^\rho(W_t \alpha_t(A_r) W_t^*) \quad (\text{A.2})$$

realize the declared standing-reconstruction witnesses for $r \in \mathcal{R}_{\text{ext}}$ and $W_t \in \mathcal{W}_{\text{ext}}$. Two representations are admissibly equivalent when all declared exterior standing functionals agree.

Definition A.4 (Horizon reconstruction sector). A horizon reconstruction sector is an equivalence class

$$[\rho]_{\partial H} = \{\rho' : \omega_{r \rightarrow w}^{\rho'}(t) = \omega_{r \rightarrow w}^\rho(t) \text{ for all admitted } r, w, t\}. \quad (\text{A.3})$$

This is a boundary-functional reconstruction-sector notion, not a DHR sector classification.

Theorem A.5 (Boundary-functional sector form). *An admissible horizon reconstruction determines a boundary-functional reconstruction sector preserving all exterior standing functionals. A representation with an inaccessible permanent interior but no induced exterior standing functional either changes the sector, is bookkeeping-inert on the exterior scope, or fails to preserve exterior standing.*

Proof. The exterior standing functionals are the algebraic representatives of $\mathcal{W}_{\text{ext}}$. If a representation preserves all such functionals, it lies in the same reconstruction sector. If it changes one, it changes the exterior witness object. If it adds interior data without exterior functional effect, the addition is bookkeeping for the declared scope. If it treats the interior data as standing-bearing without exterior functional induction, it imports a hidden selector or changes scope. Thus the AQFT-style reading agrees with the categorical obstruction. $\square$

B Horizon reconstruction degeneracy matrix

ID	Degeneracy / escape route	Blocking result
D1	Permanent horizon with no leakage or certificate	No-permanent-horizon reconstruction theorem
D2	Global information retention treated as exterior recovery	Global retention is not same-scope recovery
D3	Holographic encoding claim without anchor map	Certificate necessity
D4	Complementarity without redescription correspondence	Standing-transport certificate criteria
D5	Stable remnant as inaccessible storage	Conservative hidden-sector collapse
D6	Infinite-time recovery outside declared continuation	Scope lock and no-repair discipline
D7	Interior-only standing	Corollary: interior-only standing does not preserve exterior standing
D8	Final-state or post-selected repair	No-selector/no-repair discipline
D9	Baby universe or disconnected branch	Scope transport unless exterior certificate induced
D10	Hidden microstate label preserving all exterior witnesses	Conservative reconstruction collapse
D11	Exact classical event horizon treated as sufficient	Event horizon versus admissible reconstruction distinction
D12	Quantum-gravity completion asserted without exterior map	Route-collapse theorem
D13	Admissibility constraints treated as persistence guarantee	Exterior reconstruction non-guarantee principle
D14	Empty successor set renamed as permanent horizon continuation	No false permanence corollary
D15	Cosmological or observer horizon treated as same-scope counterexample	Scope typing and route H15 collapse

C Symbol ledger

Symbol	Meaning
$\mathcal{S}_{\text{hor}}$	Fixed exterior horizon reconstruction scope.
$\mathcal{E}_{\text{ext}}$	Exterior admissible description domain.
B_H	Candidate horizon or horizon-like boundary.
$\mathcal{R}_{\text{ext}}$	Standing-bearing exterior reference classes.
$\mathcal{W}_{\text{ext}}$	Exterior witness family.
Cont	Admissible continuation relation.
Env	Envelope ledger: physical assumptions and numerical/dynamical inputs not derived here.
X_H	Source-horizon-exterior carrier.
E_{ext}	Exterior witness envelope.
$E_{\partial H}$	Boundary/horizon trace envelope.

E_{int}	Interior or permanently isolated envelope relative to exterior scope.
T_H	Partial standing-transport operator.
Cert_H	Standing-transport certificate family.
O_H	Internal horizon totalization obstruction.
$\text{Succ}_{S_{\text{hor}}}(H^\partial)$	Standing-preserving admissible exterior successor set for a partial horizon reconstruction object.
$\text{Exh}_{S_{\text{hor}}}(H^\partial)$	Scope-indexed exhaustion of exterior reconstruction: empty successor set under fixed constraints.
$\mathcal{C}_{\text{HOR}}$	Category of finite horizon reconstruction objects.
$\mathcal{C}_{\text{HOR}}^\partial$	Partial horizon reconstruction category with explicit operator domains.
$\mathcal{Q}_{\text{HOR}}, \mathcal{Q}_{\text{HOR}}^\partial$	Universal admissible horizon reconstruction quotients.
$\Sigma(\widehat{D})$	Standing content of a conditioned admissible description.
$\mathcal{K}_{\text{AASC}}$	AASC kernel $\{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\}$.
$\mathcal{Q}_{\text{ext}}$	Exterior standing quotient $\mathcal{R}_{\text{ext}}/\sim_{\text{ext}}$.
$\Phi_W, \Phi_C, \Phi_B, \Phi_A$	Witness, continuation, boundary-accessibility, and admissibility-status profiles.

D Proof dependency graph

- G1. Coherent exterior horizon reconstruction instantiates the AASC kernel; standing is not an optional primitive layered onto black-hole physics.
- G2. Boundary-trace fixation says a horizon boundary contributes to standing preservation only where stable exterior or boundary traces are declared.
- G3. Exterior witness completion fixes the same-scope witness algebra $\mathcal{A}_{\text{wit}}$ and blocks hidden witness sectors.
- G4. Standing-bearing exterior reference requires anchor preservation under admissible continuation.
- G5. Standing conservation excludes standing sinks and standing sources.
- G6. The standing-transport relation is a partial operator with domain fixed by exterior witnesses, boundary traces, leakage, and certificates.
- G7. An uncertified permanent horizon leaves threatened reference classes outside that partial domain.
- G8. Partial-domain closure blocks same-scope totalization of standing transport on the old uncertified horizon carrier.
- G9. Nonvanishing of O_H records the obstruction internally to $\mathcal{C}_{\text{HOR}}^\partial$.
- G10. Exterior reconstruction non-guarantee distinguishes lawful continuation conditions from guaranteed realizability.
- G11. Successor exhaustion classifies an empty exterior successor set as exhaustion, not as constraint failure or admissible permanence.
- G12. $\mathcal{C}_{\text{HOR}}, \mathcal{C}_{\text{HOR}}^\partial, \mathcal{Q}_{\text{HOR}}$, and $\mathcal{Q}_{\text{HOR}}^\partial$ lock the same-scope comparison class by admissible morphisms and quotient equivalence.

- G13. Global retention is not same-scope exterior reconstruction unless it induces an exterior recoverability certificate.
- G14. Remnant storage is not exterior standing preservation unless it exposes an admissible remnant map.
- G15. Conservative reconstruction collapse removes hidden interior labels preserving all exterior witnesses.
- G16. No-reconstruction-generating structure import proves duality, performance, global endorsement, remnant indexing, or expert declaration cannot create exterior standing unless it induces a certificate.
- G17. Standing-reconstruction acquisition exhaustion proves every rescue obtains reconstruction force by certificate, selection, repair, carrier laundering, scope transport, or bookkeeping.
- G18. Route exhaustion proves every permanent-horizon route in the declared black-hole exterior scope either supplies a certificate, collapses to bookkeeping, classifies exhaustion, violates no-selector/no-repair, or changes scope.
- G19. Therefore any continuing admissible horizon model must leak, evaporate, reconstruct, or quotient standing without sink formation; otherwise the admissible classification is exterior-successor exhaustion.

E AASC dependency summary

The AASC source corpus was reviewed for the dependency architecture used here. The following source roles are used directly.

- **Minimal Conditions for Admissible Construction:** kernel non-optionality; admissibility, standing, reference, and irreversibility are instantiated by coherent construction rather than added as optional assumptions.
- **Standing–Admissibility Identity Closure:** no second same-domain classifier and no standing repair by representational change.
- **Boundary-Trace Fixation under Admissibility:** declared stable boundary traces fix roles; absence, multiplicity, instability, scalar ranking, and post-selection fail closed.
- **Constraint Architectures for Coherent Evaluability:** witness extraction, transport covariance, and no hidden evaluative remainder support exterior witness completion.
- **The AMetric Boundary:** no hidden index, metric, ranking, canonical representative, probability, typicality rule, or final-state selector has boundary authority unless fixed by the declared domain.
- **Authority-Based Legitimacy Is Impossible:** declaration, endorsement, expert status, or sovereign authority cannot generate, restore, override, or substitute for admissibility; horizon reconstruction authority is therefore admissibility-irrelevant or inadmissible.
- **Authority and Liability Without Standing:** performance, regularity, scalability, or procedural stability do not create standing; reconstruction algorithms and dual descriptions are standing-relevant only through exterior certificates.

- **Exterior standing quotient and same-scope closure:** quotient presentation removes representative-level ambiguity and fixes the exact data preserved by admissible horizon rivals.
- **Anchor, Tensor, and Skin:** exterior anchors, propagation/witness tensors, and representational skin exhaust admissibility-bearing descriptive roles; no fourth role remains.
- **A Fixed-Domain Exhaustion Theorem:** same-scope variation must alter anchor content, witness/tensor content, admissible continuation, or scope; no extra governance-bearing locus remains.
- **Closure by Exhaustion for Same-Scope Operators:** undefined old-scope operator applications cannot be made defined by same-scope extension; totalization moves exhaust into selector, repair/generator, carrier laundering, scope transport, or bookkeeping.
- **Admissible Record Construction in Physically Non-Selective Quantum Regimes:** global or nonselective retention is not a same-scope record unless it fixes standing-bearing records on the declared downstream scope.
- **Quantum Entanglement as Boundary-Level Compatibility:** boundary-level compatibility does not license hidden common carriers or same-scope witness sectors.
- **End of the Admissible Interior:** admissibility constraints are necessary but not sufficient for persistence; exhaustion is empty successor structure under fixed constraints, not temporal process, entropy drain, or constraint failure.
- **Beginning of the Admissible Interior:** beginning/interior language is licensing language rather than origin narrative; this supports the reconstructive, not ontological, reading used for horizon successors.
- **α - Ω admissible interior capstone:** the admissible interior is structurally closed as a licensed domain; no post-interior narrative or untyped successor is admitted.
- **Baryogenesis and neutrino obstruction templates:** retrieval-irreversibility, transport certificate logic, partial-operator totalization, universal quotient reconstruction, witness completion, conservative hidden-sector collapse, and route exhaustion.

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